Who does the new work? Automation and reinstatement in occupational task space

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Abstract

When automation creates new tasks, which workers perform them? Task-based models fix the answer by construction; here workers hold occupations that are bundles of tasks, and the destination of new work is an outcome. We represent a technology as a field over the two-dimensional task space of Storck and Andersson (2026). Capital takes a task where it is cheaper, and new work becomes human only where labour stays competitive and some occupation can bind it. Calibrated, the model restores work but not pay: most new tasks bind to no occupation, and freed labour re-sorts into other occupations, carrying wage pressure into work the technology never reaches. New AI and robotics firms are founded where the model places new work, and in the industrial-robot wave's own window the predicted wage pressure tracks observed occupational wage growth independently of the wage level and of the occupation's manufacturing share.

JEL codes: J23, J24, J31, O33.
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1 Introduction

Automation does not act on the labour market as a whole. It acts on particular tasks, within particular occupations. Workers accumulate human capital in specialized domains, occupations combine multiple tasks into bundles, and technologies vary in where they act, how far they reach, and whether capital can perform the exposed work more cheaply than labour. When automation displaces some tasks and creates others, the central question is not only how many new tasks appear but which workers perform them: whether the new work reaches those whose task content was displaced, and whether performing it restores their pay. Answering it means asking where new tasks appear, which occupations can take them, and whether the tasks that return carry the priced content of the tasks that left.

The task-based literature provides the canonical mechanism (Acemoglu and D. Autor 2011). Automation displaces labour from tasks that capital can perform, while reinstatement creates new tasks in which labour retains a role (Acemoglu and Restrepo 2018; Acemoglu and Restrepo 2019). In the one-dimensional framework, reinstatement raises the labour share because new tasks are placed where labour holds the comparative advantage. That is a powerful result, but it leaves an important question under-specified: which workers benefit? If labour enters as one factor, or as a small number of aggregate skill groups, reinstatement can be studied in aggregate (Acemoglu and Restrepo 2022), but not located in the differentiated structure of work.

This paper introduces the concept of a technology field: a representation of a technology as an effectiveness field over a two-dimensional task space, with a position, a reach, and an amplitude. The task space is the polar geometry of Storck and Andersson (2026), in which occupations are bundles of tasks calibrated from semantic embeddings of O*NET task statements; the angular coordinate captures domain orientation, the radial coordinate specialization depth, and depth is rewarded in socio-cognitive directions and weakly elsewhere. The present paper extends that framework to technological change. It recasts the wage structure as a price of work, a field over the space, and gives technologies a location in the same space as the work they act on; the consequences of a technology then depend jointly on where it acts and on how the price field values the work it touches. The representation separates technical exposure from economic incidence by construction, and the distinction carries the paper's results.

The central mechanism is the interaction between a technology field and the directional price of work. Capital bears a task only where it is the cheaper producer. Within the reach of a technology, the adoption margin is therefore crossed first where human work is highly priced. Automation does not simply remove occupations; it strips paid content from the bundles that define them. An occupation may remain recognizably the same while being paid for less of what it does.

Reinstatement is modeled as task creation at the boundary of the technology field. The gradient of the field identifies where automated outputs must be validated, coordinated, integrated, or translated into adjacent work. But seeded tasks do not automatically become human work. They survive as human tasks only where labour remains cheaper than capital, and they attach to an existing occupation only where that occupation already holds the priced capabilities the new task requires. The model therefore distinguishes occupational redefinition from the birth of new occupations. If the task can be absorbed by an existing bundle, the occupation is redrawn. If no occupation can bind it, the task remains unbound; where such unbound tasks concentrate, the model marks a candidate region for occupational emergence. D. Autor, Chin, et al. (2024) study the empirical counterpart, where new work has emerged and who captured it; the geometry recasts it as location relative to priced capabilities.

This changes the meaning of reinstatement. Task creation is not automatically worker-benefiting. Reinstatement in a low-priced manual or service direction can return work without restoring pay. What matters for workers is where the new tasks appear relative to the priced capabilities they already hold.

Employment effects also depend on output demand. Automation lowers unit cost where capital replaces labour. Under competitive pricing, the cost saving is passed to consumers, and demand elasticity determines whether the cheapened direction expands or contracts employment. Elastic demand draws labour into the automated direction; inelastic demand releases it. But at the adoption margin the saving is second-order, so large employment gains require elastic demand or large infra-marginal savings deep inside the technology's core. This is the spatial form of the demand mechanism of Bessen (2019) and the second-order margin of Acemoglu and Restrepo (2018).

We implement the framework with two technologies, both located from published exposure data. The first is a broad cognitive field calibrated to task-level AI exposure and located in the high-priced socio-cognitive region. The second is the industrial-robot field of Webb (2020), located from occupational robot exposure in the low-priced technical-physical west. The same calibration places two further technologies of Webb, software and machine learning, in the geometry, so four automation waves appear in one space in the order they diffused. The aim is mechanism rather than causal identification: to show how the same economy responds differently when technology acts in different regions of task space.

We also confront the model with observed occupational wage changes between 2019 and 2025. A naive directional check passes, and we show the pass is uninformative: the model's predicted wage pressure is nearly collinear with the baseline wage level, the window is dominated by the pandemic-era compression of the US wage distribution (D. Autor, Dube, and McGrew 2023), which makes the same cross-sectional prediction, and placebo technologies, including one that does not exist, clear the same check. We report this as a finding about what the cross-section can identify, not as support, and it doubles as a caution for the growing literature that correlates AI exposure with recent wage changes.

The paper makes three contributions. The first is a spatial model of technological change. It represents a technology as a field over a two-dimensional task space, the first such representation in the task-based literature. Automation and reinstatement become operations on a surface, no longer movements along a line. Whether labour keeps a task is decided pointwise by the price of work and by the capabilities an occupation already holds, rather than fixed by construction. The model turns qualitative claims about automation into spatial predictions, about where a technology strips paid work and where it seeds new work.

The second concerns the worker consequences of reinstatement: task creation restores work without restoring pay, and the paper traces why through a single spatial story. Automation strips priced content from the bundles it touches; most of the work it seeds binds to no occupation and reaches no worker; and the labour freed by stripping re-sorts into large low-priced service and manual occupations, where employment recovers and pay does not. Reinstatement of work and reinstatement of pay come apart, and the geometry locates where.

The third is empirical. The model's sharpest prediction is spatial: reinstatement seeds new work on the boundary of the technology field. Independent data on where AI and robotics firms are founded place those firms where the model seeds. The occupational wage cross-section, by contrast, reaches the model's wage prediction and cannot identify it. In the 2019–2025 cross-section, technical exposure, the baseline wage, and the pandemic compression are collinear, so occupational wage-growth regressions manufacture support for any wage-gradient mechanism, a point we make with placebo technologies, including one that does not exist. The industrial-robot wave supplies the window the cognitive wave lacks. Re-estimated in its own era, the robot field's predicted wage pressure is nearly orthogonal to the wage level, and it tracks occupational wage growth over 1999–2007 while three counterfactual fields carry the opposite sign. The correlation survives conditioning on the occupation's manufacturing share beside the wage level, so it is not the manufacturing-wide decline; import exposure varying within the manufacturing arc remains unseparated, and the causal test remains within-occupation.

The paper proceeds as follows. Section 2 introduces the spatial occupational geometry. Section 3 defines the price field. Section 4 develops the technology field, the adoption margin, displacement, reinstatement, and the bundle operator. Section 5 closes the model with employment sorting and the output-demand channel. Section 6 describes the implementation and calibration and validates the seeding prediction against firm locations. Section 7 presents the illustrative cases. Section 8 confronts the wage mechanism with observed occupational wage changes in two windows and shows what each can and cannot identify. Section 9 discusses implications and limitations.

2 The Spatial Occupational Geometry

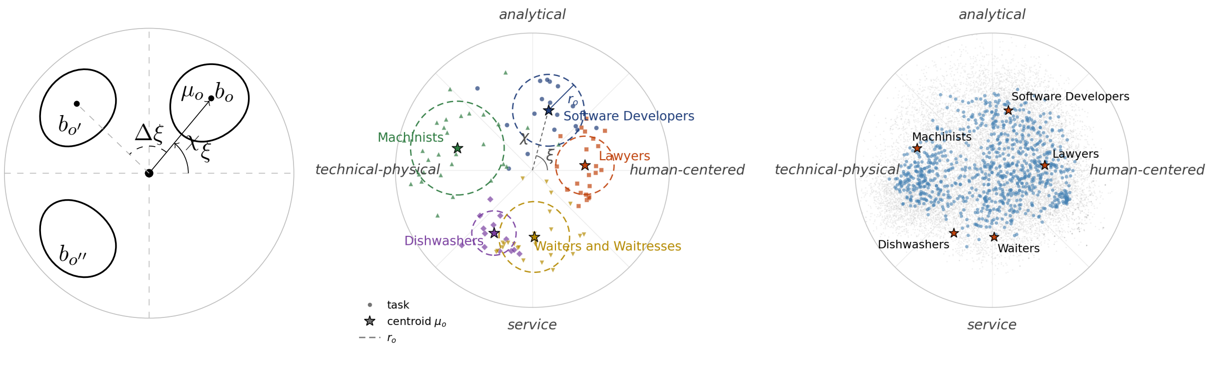
The model builds on the task-space geometry developed in Storck and Andersson (2026) where occupations are defined from their task-bundles, with all tasks distributed on a two-dimensional polar disk. Each task has a location $\mathbf{r} = (\xi, \chi)$, where ξ is the angular domain of the work and χ is radial depth or specificity. Each occupation is a weighted bundle of tasks,

$$b_o(\mathbf{r}),$$

normalized to one before technology acts. The occupation's position is given by the centroid,

$$\mu_o = \int \mathbf{r}\, b_o(\mathbf{r})\, d\mathbf{r},$$

and thus represents the point in task space where the weighted average of the task bundle is located. The coordinates are constructed in Storck and Andersson (2026): the 17,606 O*NET task statements are embedded with a sentence encoder, projected onto their first two principal components, and expressed in polar form. The orientation is frozen to a fixed reference, so positions are comparable across runs, encoders, and database releases. That paper validates the geometry against wages, education, and occupational classifications; this paper takes it as given.



(a) (b) (c)

Figure 1: The geometry of work. (a) The abstract model of Storck and Andersson (2026): occupations are bundles of tasks on the polar disk, b_o, $b_{o'}$, $b_{o''}$ three bundles with centroids μ_o; ξ is the angular domain of a centroid, $\Delta\xi$ the angular separation of two occupations, and χ the radial specificity. (b) Five occupations opened as bundles of their tasks: each point is a task, the star the bundle centroid μ_o, and the dashed circle the task radius r_o, the root-mean-square spread of the bundle around its centroid rather than an enclosing boundary; Software Developers carries the coordinate marks χ, ξ and r_o. (c) The occupational centroids of the coordinate universe (grey, 878 occupations; Section 6.1) with the four directional poles and nine example occupations; the rim is the task boundary $\chi = 1$.

The geometry matters because technologies act on tasks rather than on occupations as indivisible units. A technology may affect part of an occupation's bundle while leaving the rest intact. The relevant unit of exposure is therefore the task location, and the relevant unit of worker consequence is the rewritten occupational bundle.

The disk has a fixed compass. The east holds human-centered work: instruction, communication, advice. The north holds analytical work: research and investigation. The west holds technical-physical work: mechanical and manual execution. The south holds service work: direct service and operational support. Tasks near the origin are generic activities found in many occupations, such as organizing or documenting; tasks near the rim are domain-specific. The radial coordinate χ measures this specificity, and an occupation deep in a direction is one whose tasks are both specific and aligned.

Panel (c) of Figure 1 places nine familiar occupations on the disk. Lawyers (5°)

and Pathologists (44°) sit in the interpersonal-analytical east and northeast. Software Developers (75°) and Chemical Engineers (109°) lie in the analytical north. Machinists (164°) and Carpenters (196°) occupy the technical-physical west, Dishwashers (239°) and Waiters (272°) the service south, and Biology Teachers (340°) the instructional southeast. Throughout the paper, "the socio-cognitive north" refers to the analytical-interpersonal arc from east to north, "the manual west" to the technical-physical direction, and "the service south" to the service direction.

Panel (b) of Figure 1 opens five of these occupations and shows the tasks behind each centroid. Within a bundle the angle sorts tasks by kind. The Software Developers bundle runs from client consultation on its human-centered edge, through the analysis and system design that form its deep core, to shallow installation work reaching toward the manual west. Lawyers split into an advisory arc leaning north and a representation arc leaning toward the service side, neither radially deep: broad judgment rather than narrow specialization. Machinists span the widest and deepest arc, from service-leaning setup through a technical core of machining and diagnosis to an analytical testing edge. Dishwashers occupy a short arc of one kind, self-contained manual operations whose depth the service direction does not reward, and Waiters combine service-technical preparation with the more specific customer interaction.

The geometry also matters because the labour market does not price the space uniformly. The return to depth is directional. Depth in the socio-cognitive direction is rewarded, while depth in service and manual directions is weakly rewarded or unrewarded. This directional wage structure is estimated in Storck and Andersson (2026); Section 3 recasts it as a price field over the space. It is what allows the model to distinguish between reinstatement that restores high-priced work and reinstatement that restores low-priced work.

Because direction sorts tasks by kind, the same kind of work recurs across occupations at the same place on the disk. Conferring, analyzing and drafting sit in the core of Software Developers, in the advisory arc of Lawyers and at the coordinating edge of Machinists. Cleaning, moving and stocking sit in both service occupations. A technology defined by where it acts cuts across occupations at a location rather than along the occupational boundary, and the same technology lands in the deep core of one bundle and on the shallow edge of another. Where a task sits on the disk shapes how the market prices it. Whether a technology reaches it is a separate question, one that turns on the centre and reach of the technology, and Section 4 makes it precise.

3 The Price of Work as a Field

Storck and Andersson (2026) estimate a directional wage structure on the disk: across occupations, the return to specificity is a single first harmonic of direction, positive toward the analytical north and absent toward the service south. That is a statement about occupations, observed at their centroids. This paper lifts it to a statement about locations: the market pays a price $\Pi(\mathbf{r})$ for a unit of work performed at $\mathbf{r}$, and the wage of an occupation is the price of the work its bundle contains. The lift is what lets technologies, which act on tasks rather than on occupations, meet the reward structure of the labour market pointwise. This section constructs the field, grounds it in the capability content of the space, and states the sense in which it is an equilibrium object; Section 6.3 validates the lift from centroids to bundles.

3.1 The capability plane

The price field is grounded in a structural regularity of the occupational geometry. The capability content of work is, to first order, organized by a two-dimensional plane in capability space:

$$v_o = \bar{v} + \chi_o\Big(\cos\xi_o\, u_1 + \sin\xi_o\, u_2\Big) + \varepsilon_o, \tag{1}$$

where v_o is the capability profile of occupation o, $\bar{v}$ is the mean profile, and u_1, u_2 are the two poles of the plane. The disk is the polar representation of this plane: ξ is phase and χ is amplitude.

Under this structure, distance in task space is interpretable as distance in capability space, and capability requirements can be read as fields over the disk. For each capability cluster k,

$$q_k(\xi, \chi) = \bar{v}_k + \chi\Big(u_{1,k}\cos\xi + u_{2,k}\sin\xi\Big). \tag{2}$$

The clusters are the four directional families of O*NET descriptors identified in Storck and Andersson (2026): social-cognitive skills (S1), technical-physical skills (S2), social-cognitive abilities (A1), and technical-physical abilities (A2). S1 and A1 peak in the north and east, S2 in the west and northwest, A2 in the south and west. The market prices S1 strongly and S2 weakly; A1 and A2 carry no wage premium once the others are held fixed (Section 6.4).

The empirical implementation tests this postulate directly using O*NET capability descriptors that do not enter the construction of the task coordinates. The leading two components account for roughly three quarters of capability deviation variance and align strongly with the task disk. Higher-order structure is measured as texture and carried in sensitivity runs, not as part of the core theory.

3.2 The directional price field

If capability content is locally planar and capabilities are priced by a linear functional, then the log price of work is affine in the capability content of its location:

$$\ln \Pi(\mathbf{r}) = \kappa + \sum_k p_k q_k(\mathbf{r}).$$

Under the capability plane this implies a first-harmonic directional depth return. We carry the empirical reduced form as

$$\ln \Pi(\mathbf{r}) = m_0 + m_1 \cos\xi + m_2 \sin\xi + \chi\Big(m_3 + m_4 \cos\xi + m_5 \sin\xi\Big). \tag{3}$$

The depth interaction is the theoretical object: it is what makes depth valuable in some directions and not in others (Figure 2). The level terms capture measured structure outside the exact plane.

The coefficients are estimated once on the occupational wage cross-section and then held fixed under technology shocks. This is a deliberate partial-equilibrium choice. The technology model below asks how a given price structure mediates automation and reinstatement. It does not re-solve the full assignment of workers and tasks after the shock.

The step from occupational wages to task-location prices is testable. The field is estimated at occupational centroids, where wages are observed, but the model prices an occupation as the integral of the field over its task bundle:

$$w_o = \int \Pi(\mathbf{r})\, b_o(\mathbf{r})\, d\mathbf{r}. \tag{4}$$

Empirically, bundle-integrated wages reproduce observed wages about as well as centroid evaluation, which licenses operating the field at the task level.

3.3 Assignment interpretation

The field $\Pi(\mathbf{r})$ is used as a fixed reduced form in the main model, but it is not atheoretic. Appendix A interprets it as the price surface of a multidimensional assignment equilibrium in the sense of Lindenlaub (2017): occupations are capability types, task demand is CES over locations, and no observed wage enters the computation, so agreement with the estimated field is a test rather than a fit. Uniform demand recovers the field's shape (rank correlation +0.92) but not its directed gradient; a single cognitive demand shifter recovers both (+0.97, gradient +0.36 against the estimated +0.39). The dearness of cognitive work reads as demand-driven assignment pressure rather than productivity superiority.

The exercise supports the reduced form; the main results do not need it. The comparative statics below hold the surface fixed, and re-solving the assignment after a technology shock is a general-equilibrium extension expected to dampen, not reverse, the mechanisms studied here. The appendix also states the exercise's scope: it reproduces the field's shape, not the observed concentration of employment.

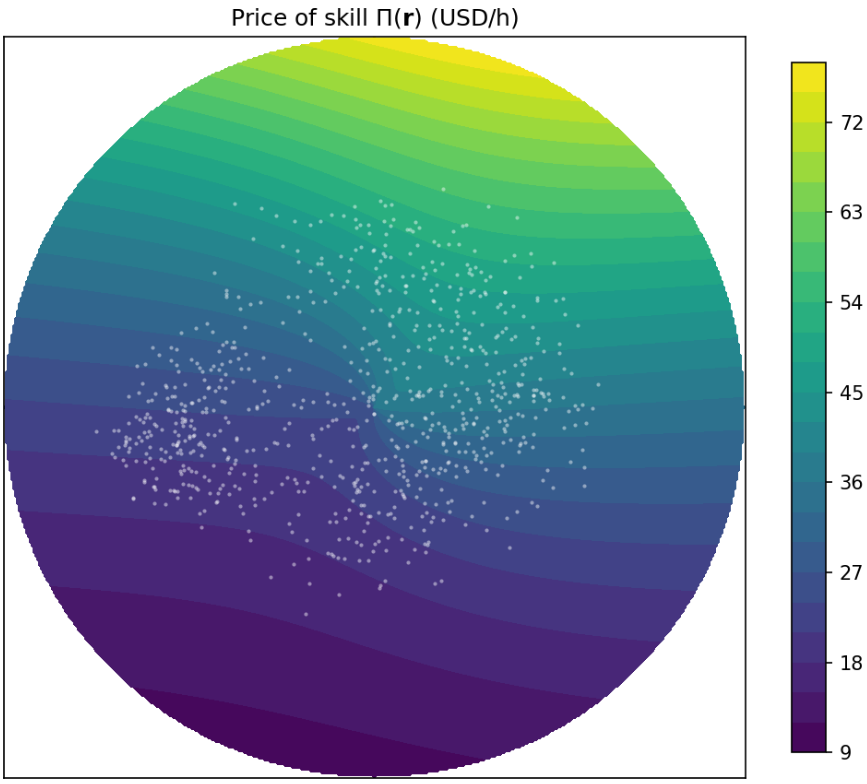


Figure 2: The price of skill $\Pi(\mathbf{r})$ over the task disk. Depth is rewarded toward the socio-cognitive north and weakly rewarded or unrewarded toward the manual west and the service south.

4 Technology Fields and Task-Bundle Rewriting

This section introduces the paper's central object: the technology field, a representation of a technology over the task space of Storck and Andersson (2026). The construction is best read against the one-dimensional model it generalises. Acemoglu and Restrepo (2018) order tasks on an interval. Capital can produce every task up to a moving threshold, and each newly created task enters at the top of the index, so displacement and reinstatement both act at the two ends of a line segment. Adoption itself is already economic in that model: an automatable task goes to capital only where doing so is cheaper. Two further steps, however, rest on the shape of the line. Comparative advantage is monotone in the task index, so the newest task is by definition the task labour is relatively best at, and a ceiling on the accumulated capital ensures that this task is performed by labour (Acemoglu and Restrepo 2018). Newly created tasks that capital would win are acknowledged and left outside the analysis (Acemoglu and Restrepo 2019).

In two dimensions, each of these devices acquires empirical content. The top of the index becomes the boundary of a technology field. The monotone comparative advantage becomes the estimated field $\Pi(\mathbf{r})$, which orders work by what the market pays for it. The capital ceiling becomes a price gate applied pointwise, so whether labour keeps a newly created task is settled location by location, and the tasks capital wins are counted rather than excluded. The line gives new work a single place to appear; the disk gives it a full circle around the field. Which occupations end up holding the new work stops being fixed by construction and becomes the outcome the model must deliver. Once occupations are task bundles, reinstatement must clear labour demand, prices, and capability constraints before it reaches a worker. That question is what the second dimension opens, and answering it occupies the rest of the paper.

A technology is represented as an effectiveness field over task space:

$$\phi_K(\mathbf{r}) = A_K \exp\left[-\frac{1}{2}\left(\frac{\|\mathbf{r} - p_K\|}{z_K}\right)^2\right], \tag{5}$$

where p_K is the centre of the field, z_K its reach, and A_K its amplitude. A narrow field represents a specialized technology; a broad field represents a more general-purpose technology.

The field does not mean that a whole occupation is automated. At each task location, the technology can bear some micro-tasks and not others. The economic question is whether capital is the cheaper producer of those micro-tasks.

4.1 The operated share

Let $s_K \in [0, 1]$ denote the replacing character of the technology. The main simulations take $s_K = 1$. Capital performs work at $\mathbf{r}$ at effective cost

$$\frac{R}{s_K \phi_K(\mathbf{r})},$$

where R is the rental cost of capital. Labour performs the same work at price $\Pi(\mathbf{r})$. Capital is the cheaper producer where

$$s_K \phi_K(\mathbf{r}) > \frac{R}{\Pi(\mathbf{r})}. \tag{6}$$

Within the reach of the technology, this condition is crossed first where labour is highly priced. The model therefore separates technical exposure, ϕ_K, from economic incidence, $a(\mathbf{r})$.

Because a location contains a continuum of micro-tasks, the condition is smoothed into an operated share:

$$a(\mathbf{r}) = \frac{1}{1 + \exp\left(-\frac{s_K \phi_K(\mathbf{r}) - R/\Pi(\mathbf{r})}{\tau}\right)}, \tag{7}$$

where τ is the width of the adoption margin. As $\tau \to 0$, the share approaches a sharp partition; for positive τ, it is the fraction of micro-tasks at a location that capital bears.

Displacement is loss of paid human task content. For occupation o, the displaced share is

$$D_o = \int b_o(\mathbf{r})\, a(\mathbf{r})\, d\mathbf{r}. \tag{8}$$

An occupation may therefore experience substantial displacement even if no spatial region is fully automated. The margin cuts inside the bundle.

4.2 Cost saving and the second-order margin

Where capital bears the operated share, unit cost falls from $\Pi(\mathbf{r})$ to

$$c(\mathbf{r}) = (1 - a(\mathbf{r}))\Pi(\mathbf{r}) + a(\mathbf{r})\frac{R}{s_K \phi_K(\mathbf{r})}. \tag{9}$$

The realized per-unit saving is

$$\Pi(\mathbf{r}) - c(\mathbf{r}) = a(\mathbf{r}) \left[\Pi(\mathbf{r}) - \frac{R}{s_K \phi_K(\mathbf{r})}\right].$$

This saving is largest deep inside the technology's core, where capital has a strong cost advantage. It vanishes at the adoption margin, because the marginal task capital just takes is one it produces at approximately the same cost as labour. This second-order margin is central to the employment results below. Displacement begins as soon as the operated share rises, but the demand channel is weak until infra-marginal savings are large.

4.3 Reinstatement as boundary task creation

Reinstatement returns a fraction γ of displaced task mass as new tasks. The question is where these tasks appear and which occupations can bind them.

New tasks are seeded where the technology field changes most rapidly:

$$\|\nabla\phi_K(\mathbf{r})\| = \frac{1}{z_K^2}\|\mathbf{r} - p_K\|\phi_K(\mathbf{r}). \tag{10}$$

This gradient peaks at distance z_K from the technology centre. Let

$$\hat{g}(\mathbf{r}) = \frac{\|\nabla\phi_K(\mathbf{r})\|}{\int \|\nabla\phi_K(\mathbf{r})\|\, d\mathbf{r}}$$

be the normalized boundary density and let

$$M = \gamma\Delta\Gamma^D$$

be the seeded task mass, where

$$\Delta\Gamma^D = \sum_o L_o D_o.$$

The seeded density is

$$\zeta(\mathbf{r}) = M\hat{g}(\mathbf{r}).$$

A seed does not automatically become human work. It survives as human work only to the extent that labour remains the cheaper producer at its location. The surviving seed density is therefore

$$\zeta(\mathbf{r})(1 - a(\mathbf{r})),$$

while

$$\zeta(\mathbf{r})a(\mathbf{r})$$

is captured by capital.

4.4 The deficit gate

A surviving seed can attach to an existing occupation only if the occupation and the task fit each other. The fit is measured in priced capabilities. The capability deficit of occupation o at location $\mathbf{r}$ is

$$\delta_o(\mathbf{r}) = \sum_{k \in \mathcal{K}} v_k \max(q_k(\mathbf{r}) - q_{o,k}, 0), \tag{11}$$

where $q_k(\mathbf{r})$ is the requirement of capability cluster k at location $\mathbf{r}$, $q_{o,k}$ is occupation o's capability content, and v_k is the corresponding component of the price functional. In the main implementation, $\mathcal{K}$ contains the priced clusters. The corresponding surplus is $\bar{\delta}_o(\mathbf{r}) = \sum_{k \in \mathcal{K}} v_k \max(q_{o,k} - q_k(\mathbf{r}), 0)$, the priced capability the occupation holds in excess of what the task requires.

Readiness to attach the task is

$$e_o(\mathbf{r}) = \exp\Big[-\Big(\delta_o(\mathbf{r}) + \lambda_{\text{over}}\, \bar{\delta}_o(\mathbf{r})\Big)/\ell\Big] \; \exp\Big(-\|\mathbf{r} - \mu_o\|/\rho\Big), \tag{12}$$

where ℓ is a global acquisition scale, λ_{over} weights the surplus, and ρ sets the attachment locality around the occupation's centroid. The form penalizes deficit and surplus together because being able to do a task and becoming its home are different questions. The deficit records what the occupation would have to acquire; the surplus records how far the task sits below the occupation's priced level, and without the penalty an over-qualified generalist far from its core would bind seed as if it were home territory. The locality term keeps attachment near the bundle the occupation actually performs.

Setting $\lambda_{\text{over}} = 0$ and $\rho \to \infty$ gives the one-sided limit

$$e_o(\mathbf{r}) = \exp[-\delta_o(\mathbf{r})/\ell], \tag{13}$$

in which only deficits bar attachment. The limit is useful for intuition — an occupation can always absorb a task that uses less of a capability it already holds — but eq. (12) is the form the model carries. Total attachment capacity at a location is

$$C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r}).$$

The bound share of surviving seed mass is

$$\Phi(C) = \frac{C}{1 + C}.$$

Occupation o receives reinstated density

$$\iota_o(\mathbf{r}) = \zeta(\mathbf{r})(1 - a(\mathbf{r}))\Phi(C(\mathbf{r}))\frac{L_o e_o(\mathbf{r})}{C(\mathbf{r})}. \tag{14}$$

The price gate and the deficit gate act in series. The first decides whether a seed survives as human work. The second decides which occupation, if any, can bind it.

4.5 The bundle operator

The post-technology bundle is

$$b_o^{\text{post}}(\mathbf{r}) = b_o(\mathbf{r})(1 - a(\mathbf{r})) + \frac{\iota_o(\mathbf{r})}{L_o}. \tag{15}$$

The occupation is therefore not abolished. Its paid task content is redrawn. Automation strips operated content; reinstatement adds bound seeded content.

The operator does not preserve bundle mass. Its integral is

$$\int b_o^{\text{post}}(\mathbf{r})\, d\mathbf{r} = 1 - D_o + B_o/L_o,$$

where $B_o = \int \iota_o(\mathbf{r})\, d\mathbf{r}$. We do not renormalize the bundle, because doing so would assume that freed time is proportionally reallocated across remaining tasks. In this model, the mass deficit is exactly the labour released to the worker-sorting layer.

The wage change implied by bundle rewriting is

$$\Delta w_o = \int \Pi(\mathbf{r}) \left[-b_o(\mathbf{r})a(\mathbf{r}) + \frac{\iota_o(\mathbf{r})}{L_o} \right] d\mathbf{r}. \tag{16}$$

The first term removes paid content, typically from high-priced tasks. The second term adds new human task content, but only where the deficit gate permits attachment. Reinstatement therefore carries a downward wage bias when it strips high-priced work and refills with lower-priced work.

4.6 Unbound tasks and candidate occupations

Not all surviving seed attaches. The unbound density is

$$u(\mathbf{r}) = \zeta(\mathbf{r})(1 - a(\mathbf{r}))[1 - \Phi(C(\mathbf{r}))]. \tag{17}$$

It collects the new tasks that survive capital but that no existing occupation can perform. The density is not an agent or a nascent occupation; it maps mismatch.

Each seeded location therefore has three possible fates:

$$a(\mathbf{r}) \quad \text{captured by capital,}$$

$$(1 - a(\mathbf{r}))\Phi(C(\mathbf{r})) \quad \text{bound by existing occupations,}$$

$$(1 - a(\mathbf{r}))[1 - \Phi(C(\mathbf{r}))] \quad \text{left unbound.}$$

These shares exhaust the seeded mass.

Where unbound mass is diffuse, it represents friction. Where it clusters, it marks a candidate region for occupational emergence. Whether such a region is an opportunity or merely low-paid residual work depends on the price field. Unbound mass in a high-Π region is latent skilled work awaiting a bearer; unbound mass in a low-Π region is closer to precarious task spillage.

5 Employment, Demand and Mobility

The task layer rewrites bundles. The worker layer determines how labour is reallocated after the rewrite. We close the model as a static spatial-sorting equilibrium and read technology effects as comparative statics between the pre-technology and post-technology rest points.

5.1 Task-work and output demand

The market structure is stated once and used throughout. Each location $\mathbf{r}$ hosts a local product assembled from its micro-task continuum in fixed proportions. Human task work at $\mathbf{r}$ costs $\Pi(\mathbf{r})$. Capital is supplied elastically at rental R and is paid its cost, $R/(s_K\phi_K(\mathbf{r}))$ per unit of task work; it retains no quasi-rent. The marginal cost of the local product is the blended index $c(\mathbf{r})$ of eq. (9), and competition passes cost changes into the product price.

The human part of occupation o's coverage is

$$h_o(\mathbf{r}) = b_o^{\text{post}}(\mathbf{r}),$$

and the capital part is

$$k_o(\mathbf{r}) = b_o(\mathbf{r})a(\mathbf{r}).$$

Total task-work at location $\mathbf{r}$ is

$$n(\mathbf{r}) = \sum_o L_o\Big[h_o(\mathbf{r}) + k_o(\mathbf{r})\Big] = \sum_o L_o b_o(\mathbf{r}) + \sum_o \iota_o(\mathbf{r}). \tag{18}$$

Two components of seeded mass stay outside eq. (18). Unbound tasks have no bearer and produce nothing. Capital-captured seeds are new capital-run production; pricing them requires re-solving the field and is deferred with the rest of the closure. Only bound seed enters the frozen-field economy. The fate accounting of Section 4.6 records all three.

With isoelastic demand of elasticity η around the baseline price, the quantity bought scales as

$$\left(\frac{\Pi(\mathbf{r})}{c(\mathbf{r})}\right)^{\eta},$$

and revenue scales as

$$\Pi(\mathbf{r})\left(\frac{c(\mathbf{r})}{\Pi(\mathbf{r})}\right)^{1-\eta}.$$

Define the demand multiplier

$$\mathcal{D}(\mathbf{r}) = \left(\frac{c(\mathbf{r})}{\Pi(\mathbf{r})}\right)^{1-\eta}. \tag{19}$$

Place output is

$$V(\mathbf{r}) = \mathcal{D}(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta}, \qquad \beta \in (0,1). \tag{20}$$

At $\eta = 1$, revenue is cost-invariant and $\mathcal{D} \equiv 1$. For $\eta > 1$, cost reductions expand revenue and draw labour into the cheapened direction. For $\eta < 1$, cost reductions reduce revenue and release labour. The congestion exponent $\beta < 1$ is local capacity; its rent $(1-\beta)V$ accrues to a fixed local factor and stays outside the wage analysis.

5.2 Occupation values

The value of occupation o is

$$W_o = \beta \int h_o(\mathbf{r})\mathcal{D}(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta-1}d\mathbf{r}. \tag{21}$$

Occupation values are coupled because many occupations cover the same task locations. When workers move into an occupation, they raise density where that occupation operates and reduce marginal value through congestion.

Human task work is paid its marginal value product, eq. (21). Pricing uses cost; payment uses marginal value. The two are not tied by a zero-profit condition here, because that condition re-solves the price field. The frozen-field model keeps this seam visible and defers the closure.

The paper carries three wage objects. The bundle wage $w_o = \int \Pi\, b_o$ prices a bundle at the frozen field and is used for cross-sectional levels. The occupation value W_o adds demand and congestion and is what workers sort on. The bundle wage change Δw_o of eq. (16) is the operator's gross re-pricing of the bundle and is the object taken to the data in Section 8. They agree on direction and differ in what they hold fixed.

5.3 Spatial re-sorting

Labour re-sorts from the pre-technology distribution. The mobility cost between two occupations is proportional to their distance in task space, task distance in the sense of Gathmann and Schönberg (2010):

$$d(j,o) = \|\mu_j - \mu_o\|.$$

The probability that a worker originating in occupation j moves to occupation o is

$$P_{j\to o} = \frac{\exp[(W_o + \alpha_o - c_m\, d(j,o))/\nu]}{\sum_{o'} \exp[(W_{o'} + \alpha_{o'} - c_m\, d(j,o'))/\nu]}, \tag{22}$$

where c_m governs mobility cost, ν logit smoothness, and the constants α_o absorb occupational attributes the value W_o does not carry: amenities, institutional wage setting, entry requirements. Employment in occupation o is then

$$L_o = \sum_j L_j^0 P_{j\to o}. \tag{23}$$

This is a fixed point because W_o depends on $n(\mathbf{r})$, which depends on the employment vector L. Total labour is fixed; allowing aggregate labour supply to respond to total value is a later extension.

The constants anchor the model to the data. They are set once, by requiring that observed employment L^0 be the fixed point of the sorting map with no technology present. Writing $v_o = \exp(\alpha_o/\nu)$, the requirement is that the destination marginals of the zero-technology transition matrix reproduce L^0, a balancing problem with a unique solution up to the additive constant that cancels in the logit; a Sinkhorn iteration recovers it. Without the constants the observed allocation is not a rest point of the logit, and a solved equilibrium mixes the technology's effect with a baseline relocation that has nothing to do with the technology. With them, $L - L^0$ is the technology's own employment response. The mobility scales come from the same baseline: ν is one standard deviation of the zero-technology occupation value, and a median-length move costs one logit unit. The sorting layer is reduced-form by design: a static allocation rule, rather than a search or matching model, that makes mobility costly in task-space distance and lets congestion close the equilibrium.

5.4 Labour share

Let

$$H(\mathbf{r}) = \sum_o L_o h_o(\mathbf{r})$$

be human task-work. The labour share of variable task-work income is

$$\Lambda = \frac{\int \mathcal{D}(\mathbf{r})\Pi(\mathbf{r})H(\mathbf{r})n(\mathbf{r})^{\beta-1}d\mathbf{r}}{\int \mathcal{D}(\mathbf{r})\Pi(\mathbf{r})n(\mathbf{r})^{\beta}d\mathbf{r}}. \tag{24}$$

The congestion exponent affects sorting but cancels from the share itself. It does not cancel from the distribution of wages across occupations: where displaced labour crowds in, the density n rises and the return $n^{\beta-1}$ falls, so re-sorting bends the effective wage down in the receiving directions and up in the vacated ones. The aggregate is silent on this because it nets to zero in Λ; the cross-section of occupation wages is not. Section 7.1 reads this second channel off the calibrated case. The demand multiplier does not cancel, because it reweights the directions whose output expands or contracts. Automation lowers the share where the operated share rises; reinstatement raises it only where new human task-work binds. Λ weights human task work by the local value of task work. It records who performs the valued work; the full income statement of a location, with capital cost, congestion rent and consumer gain, belongs to the deferred closure.

6 Empirical Implementation and Calibration

This section measures the frozen objects used by the model and calibrates the illustrative technology fields. The theory objects are the capability plane and the price functional. The measurement objects are their estimated coordinates, the bundle-pricing validation, and the family wage wedge. The simulation primitives are the technology and economy parameters varied or held fixed in the case exercises.

6.1 Data and frozen inputs

All inputs are vendored from a pinned commit of the Storck and Andersson (2026) repository. The frozen inputs are the occupation and task coordinates, O*NET task relevance ratings, OEWS median hourly wages, education levels, capability cluster intensities, and raw O*NET Skills and Abilities descriptors.

Four occupation samples appear in the paper, and each use is fixed here. The coordinate universe contains 878 occupations and is the sample of every capability-side estimation and of the geometry figures. Of these, 849 occupations carry observed OEWS employment and constitute the simulation economy of Sections 5 and 7. Of these, 785 occupations carry matched median wages and form the estimation sample of the price field and the mediation regression. Finally, 725 occupations (640 distinct SOC codes) carry a median wage in both OEWS vintages and form the sample of Section 8. The attrition is data availability at each step, not selection by the model. The task universe contains 17,606 statements, of which 17,549 (99.7 percent) join the exposure data of Section 6.5 on Task ID.

6.2 Capability plane and price field

The capability plane is estimated from descriptor deviations and tested against the task disk. The leading two components account for roughly three quarters of the capability deviation variance and align strongly with the disk coordinates. The cluster fields $q_k(\xi, \chi)$ are then estimated by OLS on occupation-level capability intensities (Table 1).

The price field is estimated by OLS of log median hourly wage on the polar regressors in eq. (3), on the 785 occupations of the wage sample. The estimation replicates the wage equation of Storck and Andersson (2026) exactly on the frozen inputs, and is printed here because the model operates on it. The estimated field, with HC3 standard errors in parentheses, is

$$\ln \Pi(\mathbf{r}) = \underset{(0.036)}{3.420} + \underset{(0.048)}{0.192} \cos\xi + \underset{(0.051)}{0.089} \sin\xi + \chi\Big(\underset{(0.090)}{-0.065} + \underset{(0.112)}{0.008} \cos\xi + \underset{(0.134)}{0.891} \sin\xi\Big), \quad (25)$$

with $R^2 = 0.523$. The isotropic depth term is indistinguishable from zero ($p = 0.47$), and the depth return is carried by a single strong first harmonic of amplitude 0.891 peaking at 89.5°, in the socio-cognitive north: depth pays only through its interaction with direction. A second-harmonic test balanced with level harmonics does not reject the single first harmonic of the depth return ($p = 0.44$); the remaining higher-order structure is mainly a level effect, not an additional depth-pricing mechanism. The surface is also not a snapshot artefact: re-estimated on the 1999, 2003, and 2007 OEWS cross-sections carried onto the same coordinates, its peak direction, isotropic term, and amplitude are stable within one standard error on a constant occupation set.[1]

Table 1: Capability plane coefficients, estimated on the full coordinate universe ($N = 878$; wages not required). Each cluster's requirement field is $q_k = \bar{v}_k + \chi(u_{1,k}\cos\xi + u_{2,k}\sin\xi)$ (eq. (2)); R^2 is the two-pole plane fit and ΔR^2 the gain from adding level harmonics and isotropic depth.

Cluster	$\bar{v}_k$	$u_{1,k}$	$u_{2,k}$	Pole	R^2	ΔR^2
S1	2.68	1.09	1.42	52°	0.74	0.001
S2	1.37	−2.22	1.23	151°	0.69	0.008
A1	3.20	0.95	1.31	54°	0.75	0.000
A2	1.57	−1.54	−0.24	189°	0.57	0.009

6.3 From centroids to bundles

The reduced-form step from occupational centroids to task locations is validated by bundle integration. The bundle-integrated wage

$$w_o = \int \Pi(\mathbf{r}) b_o(\mathbf{r}) d\mathbf{r}$$

[1]Same specification, historical wages joined through the SOC 2000–2018 crosswalks. On the balanced code set the peak direction is 93–96° across 1999–2007 against 93° on the 2023 cross-section, the isotropic term is zero in every vintage, and the amplitude lies between 0.83 and 0.96 with no trend. Full-sample fits differ across vintages, but the differences are carried by which occupation codes publish per vintage, not by repricing.

predicts observed wages about as well as evaluating the price field at the centroid ($R^2 = 0.51$ against 0.52 at the centroid; correlation 0.72). The centroid figure is the estimation's own in-sample fit, while the bundle predictor is out of specification, so parity is the informative outcome (Figure 3). The Jensen gap between the two is small (mean +0.018). This licenses the model's use of Π at task locations.

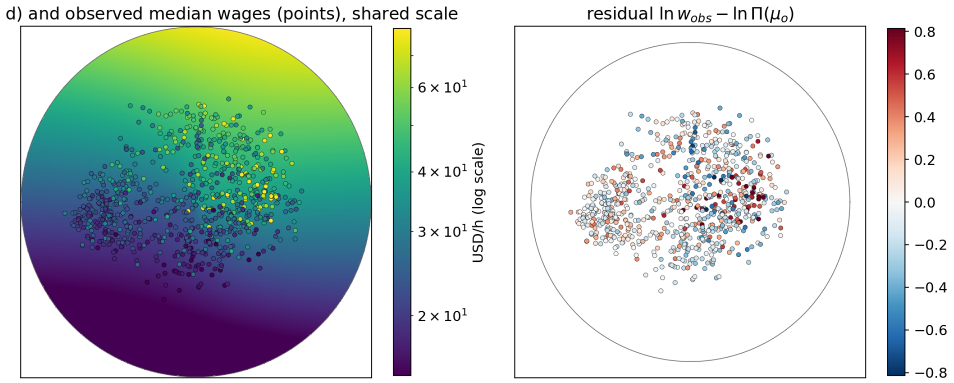


Figure 3: Bundle-integrated wage against observed wage.

6.4 Gate weights and wage wedge

The gate weights v_k are the components of the price functional. They are estimated by replicating the mediation regression of Storck and Andersson (2026) on the wage sample ($N = 785$, HC3 standard errors). The gate carries the sign restriction $v_k \geq 0$: a capability enters the deficit only if the market prices it. The estimates are $v_{S1} = 0.328$ (0.065) and $v_{S2} = 0.049$ (0.021); the ability clusters carry negative or negligible estimates and are excluded by the restriction, so the priced set is $K = \{S1, S2\}$, with S1 dominant and S2 smaller.

The residual wage wedge is

$$\omega_o = \ln w_{\text{obs},o} - \ln w_{\text{bundle},o}.$$

The family wedge ω_g is the family-level mean of this residual. It is measured once and held fixed under technology shocks. When used, it enters the adoption gate through

$$\Pi_{\text{eff}} = e^{\omega_g}\Pi.$$

The main mechanism is reported with and without this wedge (Figure 4).

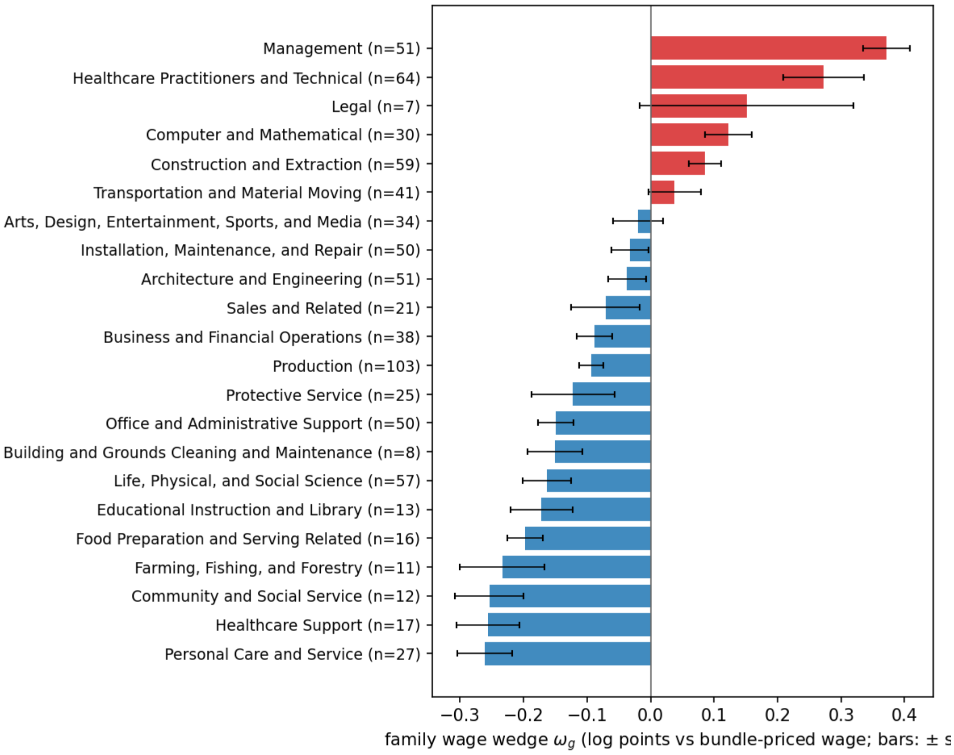


Figure 4: Family wage wedge ω_g: mean log deviation of observed wages from the bundle-priced wage, by job family. Management sits highest above the field and Personal Care and Service lowest below it; the wedge enters only the adoption margin.

6.5 Technology calibration

The cognitive technology field is calibrated to task-level AI exposure: each task statement carries the exposure score $\beta_E = E1 + \frac{1}{2}E2$ of Eloundou et al. (2024). Occupation-level exposure measures exist (Webb 2020; Felten, Raj, and Seamans 2021); the field is fitted at the task level because that is where the geometry lives. The scores are joined to the task coordinates on Task ID (17,549 of 17,606 statements, 99.7 percent) and fitted with the Gaussian field in eq. (5). The fitted field lies in the socio-cognitive north, with broad reach and moderate amplitude. The field explains a modest share of task-level exposure ($R^2 = 0.25$) but most of the smoothable between-cell exposure surface ($R^2 = 0.82$).

The modest task-level fit is a property of the rating scale, not of the field form. The exposure ratings clump at discrete values, so the between-cell share of the task-level variance — the smoothable ceiling for any field — is 0.30 on the polar grid, and the single isotropic Gaussian captures 82 percent of it. The parameters are jointly identified: the fitted centre lies inside the occupied disk ($\chi_K = 0.463$ against occupational support $\chi \leq 0.752$), so the peak amplitude is observed rather than extrapolated from a flank, and the Gauss–Newton normal matrix is well conditioned (condition number 30, with correlation -0.30 between z_K and A_K). Figure 5 shows the fitted field against the exposure surface. Directional structure the isotropic field misses is carried as texture; a richer ϕ_K is an extension, not a repair.

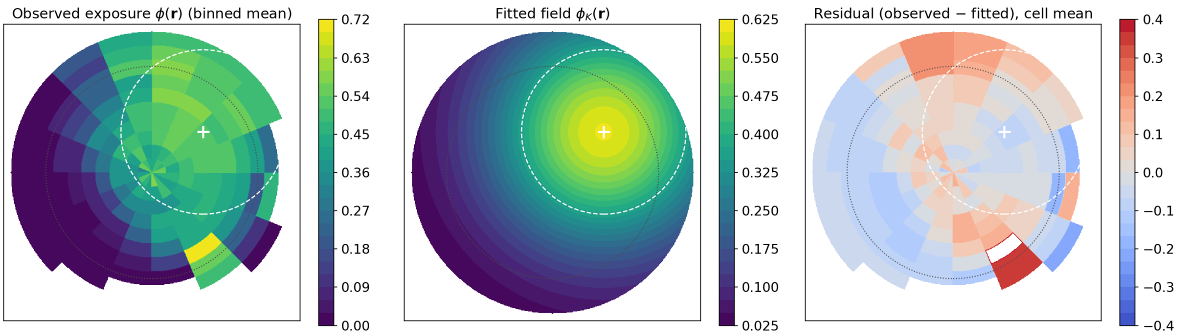


Figure 5: Task-level AI exposure surface and fitted cognitive technology field.

The economy parameters are set by explicit rules and held fixed across scenarios; Table 2 lists them. One rule is circular by construction and stated as such: the rental cost R is set so that the gate binds in the calibrated field's core, because the illustration requires an operative technology, and the R sweep of Section 7.1 carries the level of the results. The output-demand elasticity η is swept. Sensitivity of the headline quantities to these parameters is reported in Section 7.1.

Table 2: Economy parameters. Employment L is normalized to shares summing to one, so the attachment capacity C in $\Phi(C) = C/(1+C)$ is of order one. Mobility and anchoring values are set at the zero-technology baseline.

Parameter	Value	Rule
R	18	the gate binds in the calibrated field's core: capital cost $R/(s_K\phi_K) \approx 30$ at the centre against $\Pi \approx 47$; the gate closes where $s_K\phi_K\Pi < R$
τ	0.08	width of the adoption margin, eq. (7)
β	0.5	congestion benchmark; cancels from Λ
γ	0.5	seeded fraction of displaced mass; swept below
ℓ	0.133	one within-direction SD of the priced-capability index gives readiness e^{-1}
$\rho,\ \lambda_{\text{over}}$	0.5, 1	attachment locality and over-qualification weight, eq. (12)
ν	16.4	one SD of zero-technology occupation value per logit unit
c_m	31.8	the median move costs one ν
α_o	balanced	occupation constants; observed employment is the zero-technology rest point of the sorting map

Table 3: Technology calibration. The cognitive field is fitted to task-level AI exposure by least squares. Fit is the R^2 on the smoothable between-cell exposure surface. The industrial-robot field, the second technology carried through the analysis, takes its centre and reach from the Webb fit of Section 6.6; its amplitude is set in Section 7.

Field	ξ_K	χ_K	z_K	A_K	Identification	Fit
Cognitive	38.4°	0.463	0.583	0.603	calibrated	0.82

6.6 Automation waves as technology fields

The same calibration locates other technologies. Webb (2020) builds occupational exposure to three technologies by matching patent text to O*NET tasks: industrial robots, software, and machine learning. Fitting each to the disk, beside the language-model exposure, places four technologies in one geometry (Table 4, Figure 6). Webb's measures are occupation-level, so all four are fitted on the occupation substrate here; the cognitive field's centre reported below (40°) is its occupation-substrate fit, while the analysis uses its task-level calibration of Table 3.

Read in the order the technologies diffused, the fields sweep across the space. Industrial robots sit in the technical-physical west, at the periphery where production and manual work lie, with a narrow reach. Software sits to the north-west, over routine technical work. Machine learning, the AI of Webb's patents through about 2018, sits in the analytical north. Language models sit in the north-east, broad over high-priced cognitive work. The centres fall in an arc from the physical west to the cognitive north-east, in the order the waves arrived. This is the spatial form of the succession of automation technologies documented by Acemoglu and Restrepo (2020) for robots and by David H Autor, Levy, and Murnane (2003) for computerisation: each wave acts on different work, and the geometry locates where.

Two of the four fields are localised. The robot field and the language-model field centre inside the occupied disk, and their reach follows a concentrated exposure. The robot field is narrow and western; the language-model field is broad and north-east. The software and machine-learning surfaces are gradients rather than peaks. Exposure rises toward the north without an interior maximum, so a single field centres at the edge of the occupied region and its reach is an artefact of that shape. Webb's machine learning locates AI further north than the language-model measure and does not coincide with it, which is expected, since patent-era machine learning and language models are different technologies acting on different work.

Three limits bound the reading. The chronology is external to the data. Webb's three measures are built from patents through about 2018, so they do not date the technologies; the order comes from when each diffused. The measures do not share a source, since robots, software, and machine learning are one construction and language models another.

And four fields illustrate the mechanism, not a temporal law. Within those limits the figure does what the representation is for: it places several technologies in one geometry according to where each acts.

Table 4: The four technology fields fitted on the common occupation substrate, in the order the technologies diffused. Robots, software, and machine learning are the patent-based exposures of Webb (2020); language models are the LLM exposure of Eloundou et al. (2024). Fit is the R^2 on the between-cell exposure surface; the centre column marks fields whose centre sits at the edge of the occupied disk (support $\chi = 0.75$). The cognitive field's analysis calibration is the task-level fit of Table 3.

Technology	Source	ξ_K	χ_K	z_K	R^2	Centre
Industrial robots	Webb	207°	0.75	0.50	0.84	at edge
Software	Webb	146°	0.75	0.80	0.62	at edge
Machine learning	Webb	108°	0.75	0.70	0.70	at edge
Language models	Eloundou	40°	0.49	0.57	0.83	localised

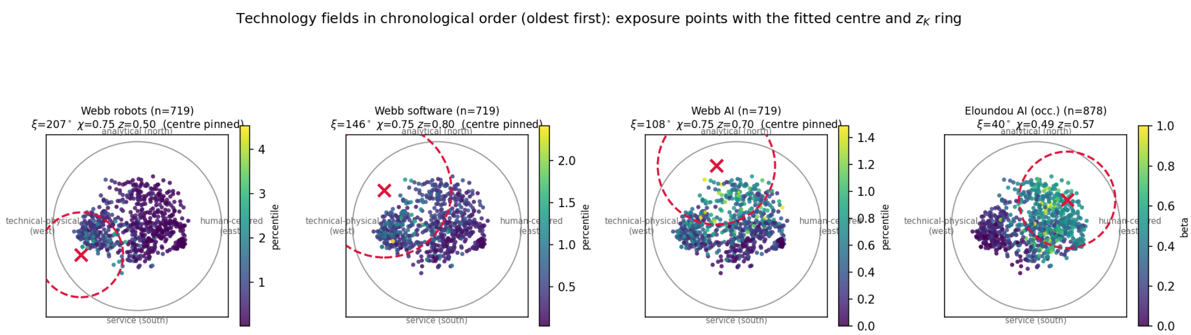


Figure 6: The four technology fields, oldest first. Each panel shows the occupations coloured by their exposure to one technology, with the fitted field centre ($\times$) and the z_K ring. Industrial robots, software, and machine learning are from Webb (2020); language models from Eloundou et al. (2024). The robot and language-model fields enclose a concentrated exposure; the software and machine-learning surfaces rise toward the north without an interior peak, so their fitted centre sits at the edge of the occupied disk.

6.7 External validation: where new firms appear

The calibration can be confronted with firm locations it never used. Reinstatement seeds new tasks where the technology field changes most rapidly, on the ring at radius z_K where $\|\nabla\phi_K\|$ peaks (10). This is the two-dimensional form of an older claim. In the one-dimensional model, new tasks enter at the top of the task index, above the range capital has taken, where labour holds the comparative advantage (Acemoglu and Restrepo 2018; Acemoglu and Restrepo 2019). The field lifts that top to a boundary. New work should appear at the boundary, away from the technology centre, and a scalar exposure ranking has no boundary and no gradient, so it cannot place new work at all.

The validation needs firm locations fixed without reference to where the model seeds. We take the AI and robotics startups of Fenoaltea et al. (2026), tagged from the Y Combinator directory ($n = 1894$ and 155; 78 carry both tags). Each firm's product text is embedded with the encoder of Storck and Andersson (2026) and mapped through the frozen projection, which was calibrated to O*NET task statements and to wages and never to where firms are founded. A control corpus of 1,500 companies from the same directory carrying neither tag, embedded and projected identically, supplies the empirical null: whatever placement generic startup text receives, the technology groups must exceed it. Document embeddings shrink norms toward the corpus mean, so radial positions are conservative; angular placement carries the group separation and is unaffected.

The two populations take distinct positions, and the ring lies between them (Figure 7). The AI firms sit on the ring's inner flank: median distance 0.45 to the technology centre against the control's 0.49, with disjoint bootstrap intervals, a compass shifted toward the analytical north, and incidence-field density 0.92× the disk mean against the control's 0.30×, three times more capability-near than generic startup text, on the side of the boundary where the model binds new work to existing occupations. The robotics firms sit on the outer western arc: 65 percent in the technical-physical west against the control's 11, out of the incidence core entirely (0.03×), under the arc the model keeps unbound. Raw radial statistics do not carry the comparison: any document corpus shrinks toward the centre of the projected cloud, which itself sits near the ring radius, so the control matches the disk-null ring enrichment (1.33× against the technology groups' 1.30 and 1.34) and the discrimination runs on structure. A cross-section of this kind cannot confirm the prediction; it could have contradicted it, with populations in the core or where the control sits, and does not.

Bound reinstatement does not spread evenly around the ring. It concentrates in two lobes, one near the technology centre and one in the technical-physical west, which appear as two peaks in the radial profile of B_o (Figure 7). The two technologies fill a lobe each, and the split follows from where each sits in the geometry rather than from a second mechanism. AI firms sit on the inner lobe, where existing occupations can bind the new work. Robotics firms sit on the outer lobe, off the incidence core entirely, where the seeded work binds to no existing occupation. A scalar exposure ranking scores the two alike. Pooled, the firms trace both peaks of the model's bound reinstatement.

7 Illustrative Cases

We run the cognitive field through the calibrated economy, and add the industrial-robot field in the demand channel. The robot field's centre and reach are the Webb fit of

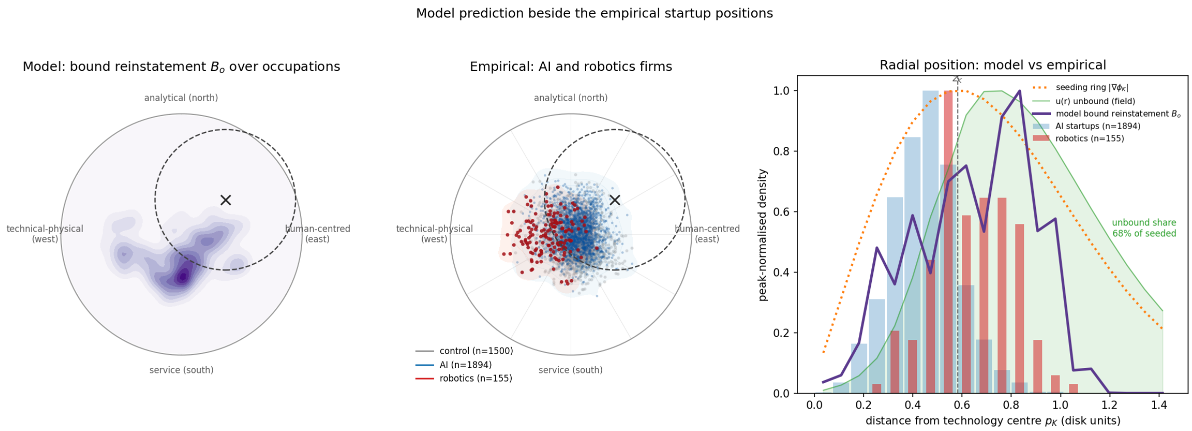


Figure 7: The model's prediction beside the firm positions, on the same disk. Left: bound reinstatement B_o over occupations, concentrated on the seeding ring at radius z_K (dashed) around the technology centre p_K. Middle: AI firms (blue) and robotics firms (red) from Fenoaltea et al. (2026), placed by their product text through the frozen projection; grey points are the control corpus. Right: radial density against distance from p_K; the seeding ring $\|\nabla\phi_K\|$, the unbound field $u(\mathbf{r})$, the model's bound reinstatement, and the AI and robotics firms all peak near z_K. The unbound share of seeded mass is 68 percent. The firm positions are calibrated to O*NET and wages, never to where firms are founded.

Section 6.6. Its amplitude is a modelling rule of the same kind that sets R: $A_K = 1.2$ opens the gate moderately in the west, and the same constants carry through the placebo of Section 8. A calibration of the gate to the aggregate displacement of Acemoglu and Restrepo (2020), reported in Section 8, returns $A_K = 1.48$ at the 1999 price field, the same order as the rule. The purpose is illustrative. The cases show what the geometry makes visible: where work is taken, where reinstatement is seeded, which occupations can bind it, how much remains unbound, and how demand elasticity changes employment consequences.

7.1 The cognitive field through the equilibrium

The cognitive field operates the disk according to the price gate. Within bands of equal technical effectiveness, the operated share orders locations by price (the within-band rank correlation $\rho(a, \Pi) = 0.99$ is a mechanical property of the takeover rule, not independent evidence). Capital therefore moves first against dear work within the exposed region (Figure 8).

Solving the worker layer before and after bundle rewriting gives the comparative static. The labour share falls substantially under pure automation, from a pre-technology normalization of one to 0.65, and reinstatement does not restore it, leaving it at 0.644 in the main calibration. Task creation is not absent; much of the seeded mass either fails to attach or attaches in lower priced regions. The geometry withholds the aggregate recovery that a one-dimensional model would assign to labour by construction. Aggregate

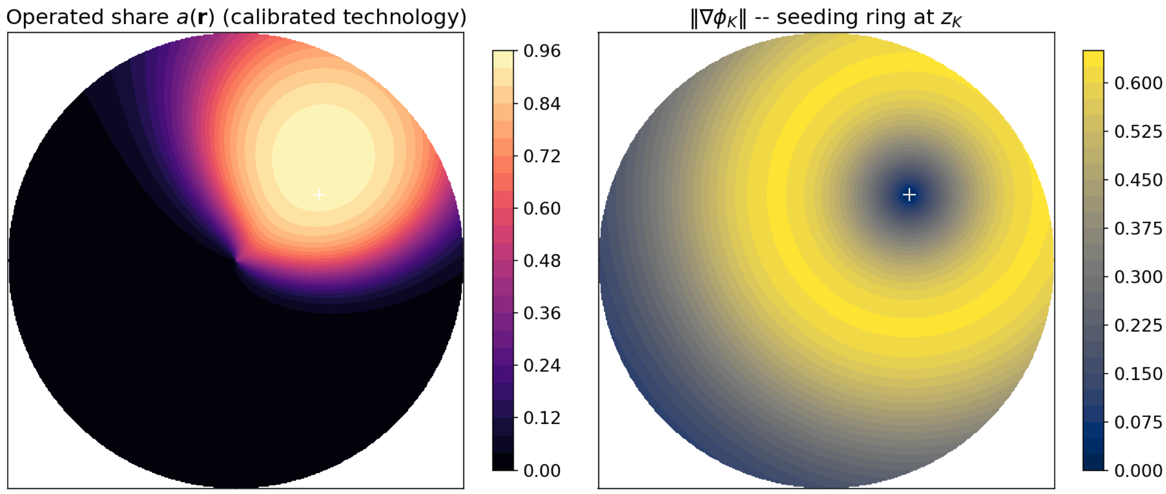


Figure 8: Operated share and reinstatement boundary for the calibrated cognitive field.

employment does not fall, because total labour is fixed and re-sorts; the comparison is carried by pay and by who performs which work. The result is robust to the family wage wedge: with the wedge in the adoption gate, the share falls to 0.648 under automation and 0.643 after reinstatement, the unbound share of seeded mass is unchanged at 68 percent, and the ordering of gaining and losing occupations is preserved.

The level of the automation drop is set by the gate. Sweeping R from 12 to 24 moves the post-automation share from 0.51 to 0.78, because R fixes how deep the technology cuts. The structure of the result is not. Across a sixteenfold range of the attachment scale, γ from 0.25 to 0.75, both margin widths, and the attachment shape, reinstatement moves the share by less than one percentage point in either direction, against an automation drop of 22 to 49 points; at the most generous corner it adds 0.4 points. The unbound share of seeded mass stays between 53 and 79 percent over the full sweep, 61 to 72 within the attachment sweep. Re-sorted mass runs from 9 to 19 percent, set by the depth of the cut: cheap capital cuts deeper and frees more labour. The claim the paper rests on is the gap between task creation and wage recovery, not the level of the drop.

The shock relocates 14 percent of employment mass. Losers are exposed occupations whose high-priced bundle content is stripped: clinical nurse specialists, sustainability specialists, software developers. Gainers are the large low-priced service occupations that absorb the freed labour: retail salespersons, baristas, fast-food workers, cashiers. Binding also pulls employment toward occupations able to perform the seeded tasks (the employment response correlates with bound reinstatement at +0.72), but the absorption in levels sits in low-barrier service work. The manual arc gains 0.04 of employment mass as displaced labour re-sorts there, without a corresponding wage recovery because depth is weakly priced in that direction.

The re-sorting that absorbs the freed labour has a wage consequence of its own, and the analysis is incomplete without it. Existing occupations take in the displaced mass by growing, and because local capacity is congested ($\beta < 1$), growth lowers the marginal return in the receiving occupations. Decomposing each occupation's wage change exactly into the content stripped from its bundle and the congestion induced by inflow, stripping carries the mean: an employment-weighted -0.35 log points, against a congestion mean near zero (-0.03). Congestion is a distributional channel. It moves a mean absolute 0.10 log points across occupations, twenty-nine percent of stripping's magnitude, with the sorting signature: occupations that gain employment crowd themselves (-0.06 on average), and the thinned-out exposed occupations recover part of their loss ($+0.22$). The level is a full-maturity re-pricing rather than a forecast. Figure 9 maps the channel. The cognitive field sits in the north, the freed labour re-sorts toward the least-damaged parts of the structure, which are largely manual and production directions, and the effective wage field is bent down where it lands. The model predicts that cognitive automation transmits a modest wage pressure into the production and manual occupations that absorb the freed labour, through re-sorting rather than through the technology's reach. This prediction is directional; the cross-section of Section 8 cannot identify it, and the channel is consistent with the paper's other occupation wage objects.[2]

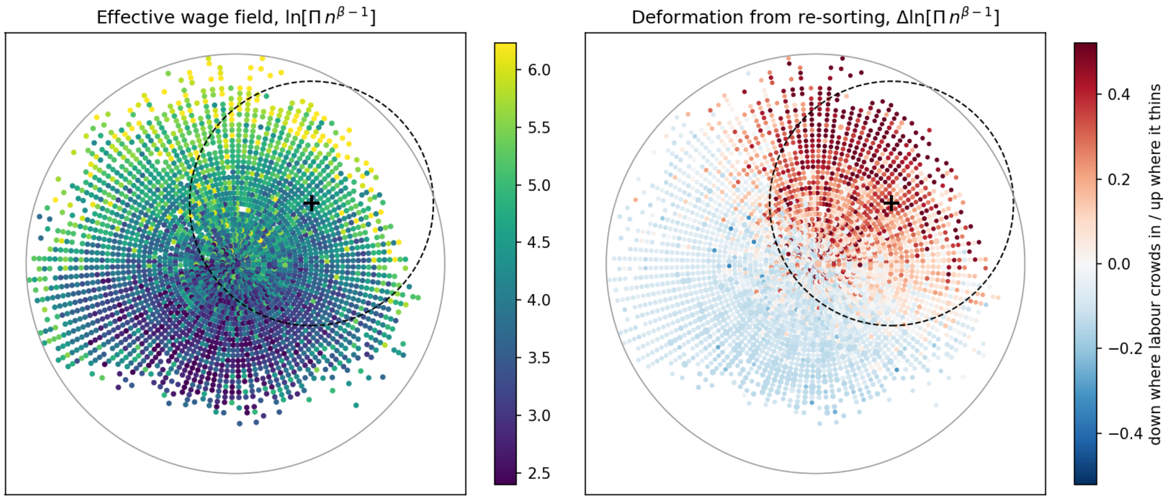


Figure 9: The effective wage field $\Pi(\mathbf{r})\, n(\mathbf{r})^{\beta-1}$ after re-sorting (left) and the wage pressure re-sorting induces (right), over the task disk. Freed cognitive labour re-sorts toward the least-damaged directions, largely the production and manual arc, where crowding bends the effective wage down (blue). The technology field is centred in the socio-cognitive north (cross); the burden falls on occupations it does not reach.

[2] The stripping-only bundle wage change, the first-order directional pressure of Section 8, and the congestion-inclusive value change agree in sign against observed wage growth and rank occupations consistently (Spearman $+0.84$ between the two stripping measures and $+0.91$ once congestion is added, the congestion channel being small); adding congestion does not move the correlation with the confounded 2019–2025 cross-section, as it should not.

7.2 Unbound tasks

Of the seeded mass in the cognitive case, 27 percent is captured by capital, 5 percent binds to existing occupations, and 68 percent survives capital but finds no bearer under the deficit gate. Of the seed that survives as human work, 93 percent is unbound. Its spatial distribution marks the gap between task creation and occupational absorption (Figure 10).

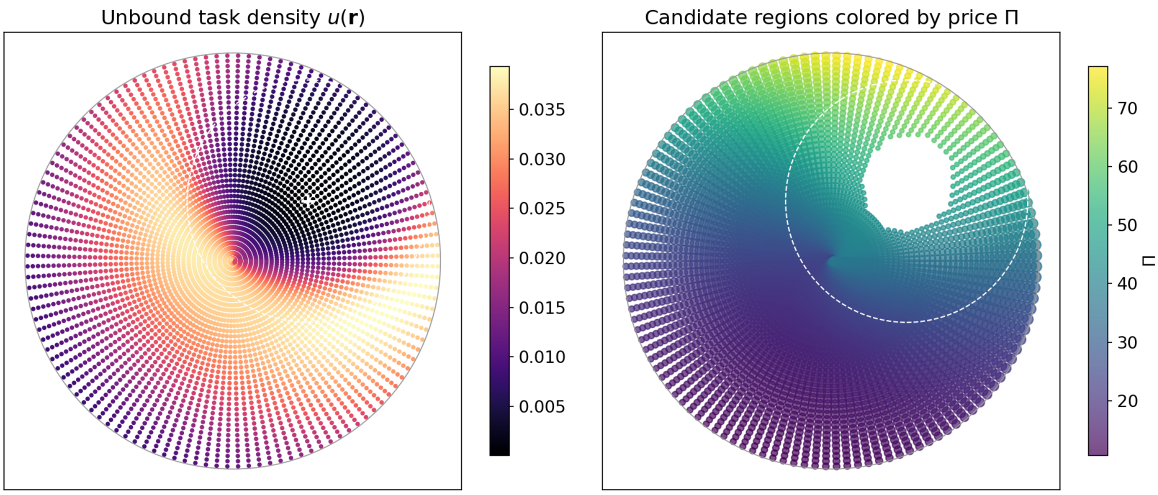


Figure 10: Unbound task density and candidate regions for occupational emergence.

The reading follows Section 4.6: whether this mass is latent skilled work or low-paid friction depends on the price at its location. The model does not create new occupations; it identifies where they may become necessary.

7.3 The demand channel

The demand channel is evaluated by sweeping the output-demand elasticity η. At unit-elastic demand, cost savings leave revenue unchanged. Both technology fields release labour in their automated region, by 12 percent for the cognitive field and 22 percent for the robot one (Figure 11). Across plausible elasticities, displacement dominates because savings are small at the adoption margin: the price-to-cost ratio is essentially one there against roughly 1.5 deep in the cognitive core and 1.2 in the robot core. Net employment growth requires elasticity beyond the plausible range: the cognitive region crosses zero at $\eta \approx 8$, and the robot region does not cross below $\eta = 12$, because its core saving is smaller and the demand channel has less to hand back.

This recovers the contrast emphasized by Bessen (2019). Automation in an inelastic-demand direction releases labour; automation in an elastic-demand direction can draw labour in. The disk adds the location. Mechanised agriculture and nineteenth-century textiles occupy different places on the elasticity scale; cognitive automation is conditional on the still uncertain elasticity of demand for cognitive output.

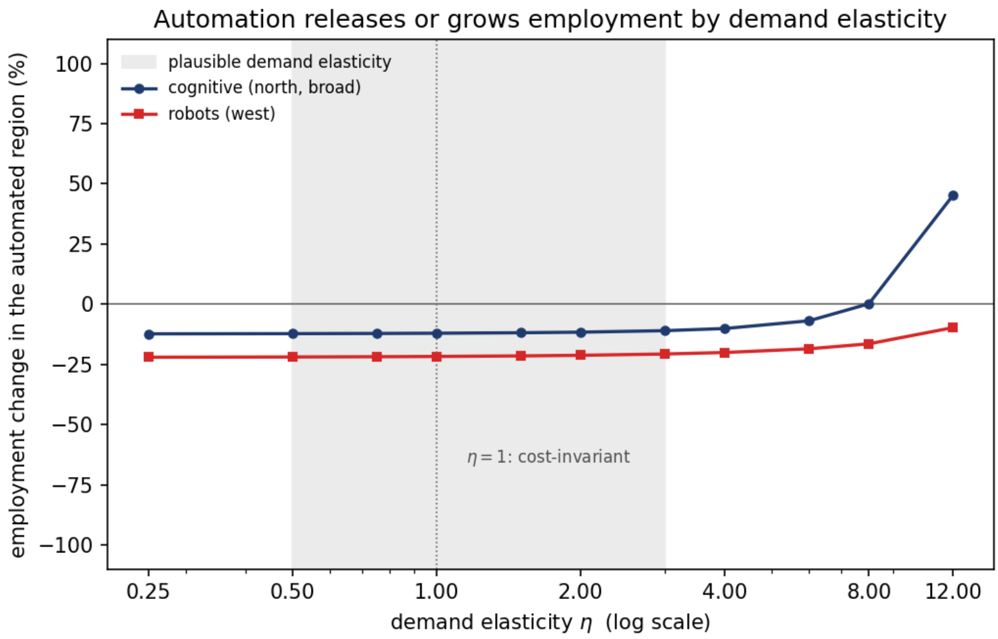


Figure 11: Employment change in each technology field's automated region as a function of output-demand elasticity.

8 Confronting the model with data

The model's wage claim is about incidence, about where automation strips paid content and pushes wages down. Observed occupational wage changes bear on it in two windows, the cognitive wave's own and the industrial-robot wave's own, and the two speak with different authority. We take each in turn and then say what each can and cannot show, beside the location validation of Section 6.7.

8.1 The naive directional check

The wage claim is directional. If automation and reinstatement push an occupation's bundle down the price gradient, that occupation should experience weaker wage growth than occupations whose bundles move less or remain closer to high-priced work. We confront that implication with observed occupational wage changes in two steps, a naive directional check, which the model passes, and a confound analysis, which shows the pass is uninformative in this window. The second step matters beyond this paper.

Using the calibrated cognitive field, each occupation's bundle is rewritten and its centroid shifts from μ_o to μ_o^{post}. The predicted directional wage pressure is

$$\widehat{\Delta \ln \Pi_o} = \Delta\mu_o \cdot \nabla \ln \Pi(\mu_o). \tag{26}$$

A negative value means that the bundle moves down the price gradient. Because the prediction is negative for almost every occupation, a *positive* rank correlation with wage growth means that occupations with more severe down-gradient shifts see lower wage growth.

We compare this predicted pressure to observed OEWS median hourly wage changes between May 2019 and May 2025 (the latest OEWS release), matched by SOC code; 725 of 849 occupations (640 distinct SOC codes) carry a median wage in both vintages. Because the match is many-to-one and the wages nominal, we compare ranks, which differences out the common nominal drift (mean $\Delta \ln w \approx +0.22$). The field pushes almost every bundle down-gradient (96 percent), and the naive check passes: the rank correlation between predicted pressure and observed wage change is +0.34 across occupations ($p < 10^{-20}$) and +0.35 at the SOC level; the model's own bundle wage change tracks observed changes slightly better (+0.42), and the result is stable when May 2024 is used as the endpoint (+0.39).

8.2 The confound

This looks like support for the mechanism. It is not. The predicted pressure is nearly collinear with the baseline wage level: because the price gate targets dear work, occupations with high 2019 wages are pushed furthest down-gradient (rank correlation −0.57 between predicted pressure and baseline wage). And the window contains the pandemic-era compression of the US wage distribution, in which low-wage occupations saw exceptional relative wage growth for reasons unrelated to automation (D. Autor, Dube, and McGrew 2023): baseline wage alone predicts wage growth at rank correlation −0.58. The model and the compression therefore make the same cross-sectional prediction, and the raw correlation cannot distinguish between them.

The data resolve the collinearity in favour of the confound. Controlling for the baseline-wage rank, the partial rank correlation between predicted pressure and wage growth is +0.02 ($p = 0.57$); the bundle wage change performs no better (+0.03). Adding the mechanism raises the rank-R^2 already achieved by baseline wage alone by 0.0003. The fine structure points the same way: the wage-growth gradient is steepest where the field barely reaches and nearly flat where it is strongest, and among above-median-wage occupations, where the mechanism strips the most and the compression matters least, the raw correlation vanishes entirely (−0.03). This is compression at the bottom of the distribution, not high-priced content stripped in the exposed region.

A window split settles the matter. The OEWS 2023 vintage divides the period into 2019–2023, which contains the peak of the compression and essentially predates the deployment of the technology the field is calibrated to, and 2023–2025, the period of actual diffusion. Table 5 and Figure 12 report the result: the entire raw correlation lives in the pre-deployment window, and in the deployment window it is absent, along with most of the compression gradient itself.

A placebo makes the machinery explicit. We repeat the exercise with two technologies that make no claim about the period's wage changes: the industrial-robot field of Section 6.6, a real technology whose deployment predates the window (Acemoglu and Restrepo 2020), and a rotated clone of the cognitive field, a technology that does not exist. The robot field returns the null. Its predicted pressure loads only weakly on the wage level (collinearity -0.09 against the cognitive field's -0.57), so the compression hands it nothing: the raw correlation is $+0.04$ ($p = 0.25$), gone under conditioning and absent in every window (Table 5). The rotated clone, sitting mostly in cheap territory where the gate barely opens, loads on the wage level at $+0.22$ and collects a weak but significant correlation (-0.13, $p = 4 \times 10^{-4}$), null given baseline wage and confined to 2019–2023. The two placebos bracket the mechanism. What the naive check finds is set by how strongly a footprint loads on the wage level, not by the technology's relevance to the period. A real technology whose wave is over returns the null. A nonexistent one that loads on the wage level still clears conventional significance. The cognitive field, loading hardest, is the $+0.34$ that vanishes under conditioning.

The robot field also shows what the geometry can say where the wage data cannot. Its reinstatement seeds on the western ring, in the technical-physical arc, where the price field is low and the priced capabilities the seeded tasks require are scarce. Run through the full equilibrium of its own era, with the price field re-estimated on the 1999 wage cross-section and the gate scale calibrated so that displaced task mass matches the aggregate displacement estimated by Acemoglu and Restrepo (2020), the field turns the implication into a number: 93 percent of robot-seeded mass binds to no occupation, against 68 percent for the cognitive field, stable between half and double the calibration moment and at observed employment. The bound residual lands on the technician arc, in the occupations that integrate and maintain the machines. Robot reinstatement returned almost no offsetting employment, consistent with their finding of no positive employment effects outside the exposed region.

8.3 The robot wave's own window

The industrial-robot wave supplies what the cognitive window lacks. Its diffusion years are covered by a classification-consistent wage window, 1999 to 2007 with 2003 as midpoint, frozen onto the geometry through composed SOC crosswalks, and the confound of Section 8.2 is absent by measurement: the robot field's predicted bundle wage pressure at Π_{1999} loads on the 1999 wage level at $+0.03$, so the wage level explains none of whatever correlation appears.

The instrument is the bundle wage pressure Δw_o rather than the directional projection, and the geometry forces the choice. The projection records the direction a residual centroid moves. A field in the dear region strips dear content and points residuals down the gradient;

Table 5: Window split of the directional check. Spearman rank correlations between predicted directional wage pressure and observed wage change; the partial correlation controls for the rank of the wage level at the start of each window. The baseline gradient is the rank correlation between the starting wage level and wage growth ($N = 725$). The placebo panel repeats the exercise with the industrial-robot field of Section 6.6, a technology whose deployment predates the window; its pressure loads only weakly on the wage level, so its raw correlation is near zero.

Window	Raw	Partial | baseline wage	Baseline gradient
Calibrated cognitive field			
2019–2023 (pre-deployment)	+0.34	+0.03	−0.56
2023–2025 (deployment)	+0.04	−0.03	−0.11
2019–2025 (full)	+0.34	+0.02	−0.58
Placebo: robot field			
2019–2023	+0.03	−0.02	−0.56
2023–2025	+0.03	+0.02	−0.11
2019–2025	+0.04	−0.01	−0.58

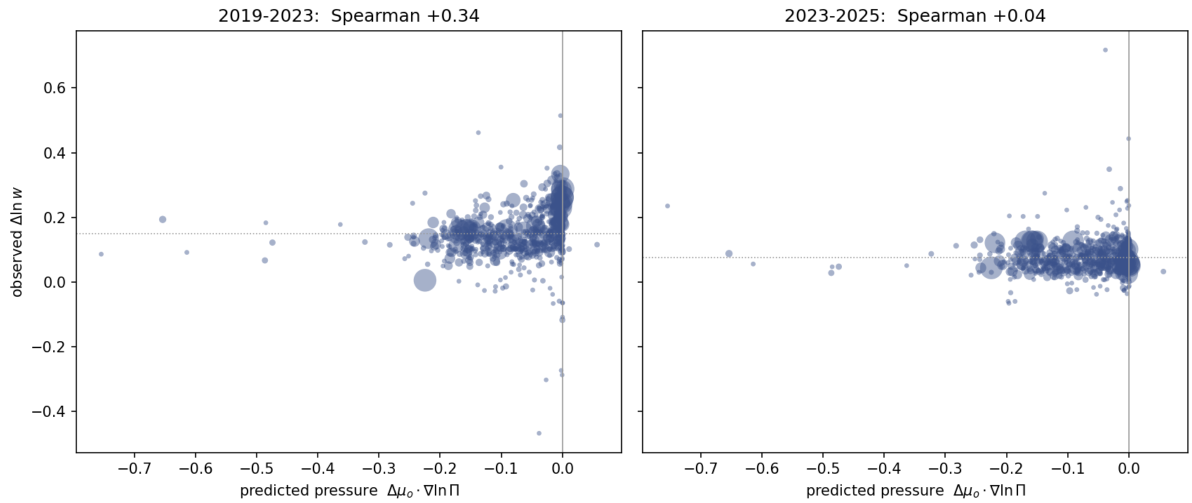


Figure 12: Predicted directional wage pressure against observed occupational wage change, split at the OEWS 2023 vintage. The raw correlation is confined to the 2019–2023 window, which contains the peak of the pandemic wage compression and predates the deployment of the technology the cognitive field is calibrated to; in the 2023–2025 window it is absent.

a field in the cheap region strips cheap content and points residuals up the gradient; and for any single-peaked field the projection is monotone in position along the gradient axis. In a window whose wage growth carries a positional trend, the projection therefore correlates for every field, real or fictitious, with the sign set by orientation alone. The bundle pressure has no such degeneracy: more stripped value predicts lower relative wage growth wherever the field sits.

Table 6 reports the result. Occupations under deeper predicted robot pressure saw lower wage growth over 1999–2007 (Spearman +0.22, unchanged at +0.22 given the baseline wage, and +0.21 on the stable-crosswalk subsample), in both halves of the window (Figure 13). Three counterfactual fields, each calibrated to the same aggregate displaced mass so that the comparison is between locations rather than scales, carry the opposite sign: the cognitive field evaluated before the technology existed (−0.29 given the wage level), a rotated robot clone (−0.22), and the software field (−0.12). Their correlations are the period's positional growth trend, the cognitive north-east pulling ahead of the wage level, and their sign is mechanism-inconsistent: their nominally pressed occupations grew faster. The robot field alone carries the mechanism's sign.

Running both waves through both windows closes the design. In 2019–2025 the objects behave as Section 8.2 found: the cognitive raw correlation (+0.42) collapses given the baseline wage (+0.03), and the robot raw (−0.20) is manufactured by a +0.31 loading on the wage level and collapses the same way (−0.02). One cell of the four survives conditioning with the mechanism's sign, the robot in its own window. The scope is that of any occupational cross-section: robots and import competition pressed the same western manufacturing arc over these years (David H. Autor, Dorn, and Hanson 2013). A manufacturing-share control meets the confound directly. Each occupation's manufacturing employment share is strongly collinear with the predicted pressure (rank correlation −0.63), and the robot correlation survives conditioning on it beside the wage level, at +0.20.[3] The correlation is therefore not the manufacturing-wide decline. What the window still cannot separate is import exposure varying across industries within the manufacturing arc, so the attribution caveat narrows without closing.

8.4 What the data show

The confrontations are not symmetric. The 2019–2025 wage cross-section cannot test the mechanism: the model's wage prediction and the dominant wage shock of the period load on the same index, the baseline wage, and once that index is held fixed no identifying variation is left. That cross-section fails to contradict the model because it cannot discriminate. The

[3]The share is built from the BLS industry-specific occupational employment files: employment in NAICS 31–33 over employment across all industries, per SOC-2000 occupation, May 2003 (the window midpoint), mapped onto the frozen geometry through the composed crosswalks. The 2002 vintage gives a rank correlation of 0.98 with the 2003 shares.

Table 6: Each wave in each window. Spearman rank correlations between predicted bundle wage pressure and observed occupational wage growth; the partial correlation conditions on the rank of the start-of-window wage level. Within each window both fields are calibrated to the same aggregate displaced task mass, set by the window's own wave ($N = 774$ in 1999–2007, 725 in 2019–2025).

	1999–2007		2019–2025	
Field	Raw	Partial	Raw	Partial
Industrial robots	+0.22	+0.22	−0.20	−0.02
Cognitive	−0.38	−0.29	+0.42	+0.03

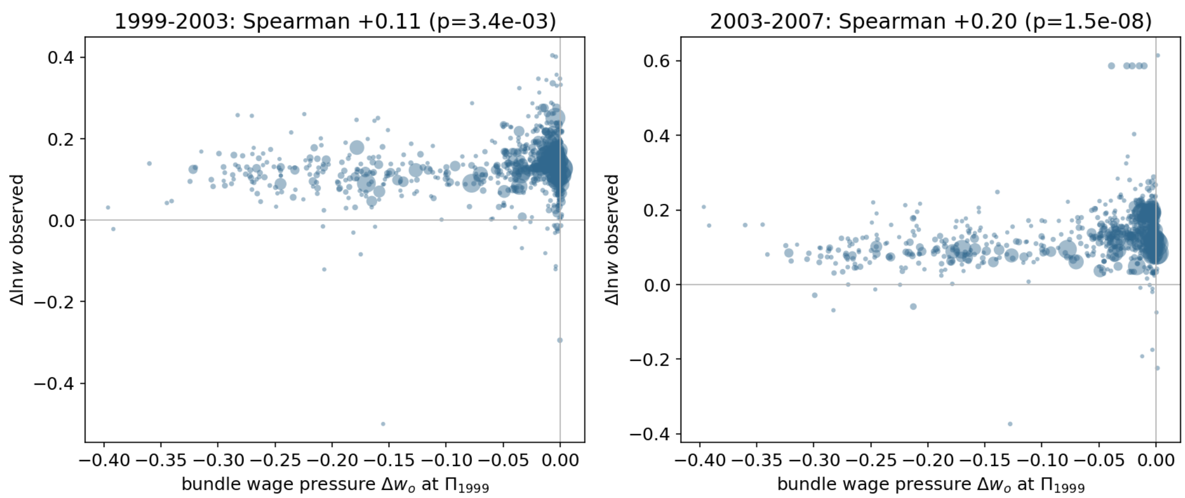


Figure 13: Predicted bundle wage pressure of the calibrated robot field at Π_{1999} against observed occupational wage growth, in the two halves of the robot window. Marker size is 1999 employment. The pressure measure is nearly orthogonal to the wage level, so the raw and conditioned correlations coincide.

robot window can discriminate, and it reads in the model's favour: the pressure measure is orthogonal to the wage level there, the correlation survives conditioning on the wage level and on the manufacturing share, and the counterfactual fields carry the opposite sign. What it cannot do is attribute, since robots and import competition pressed the same occupations. The firm locations (Section 6.7) are different again. They bear on a spatial prediction the wage data cannot make, that new work appears at the technology's boundary, and the two populations sit on the ring's two flanks where the control corpus does not. None of the three readings is proof. A single cross-section of firm positions cannot separate seeding on the ring from other processes that would deposit new work there, and an occupational wage window cannot separate contemporaneous pressures on the same arc.

The wage mechanism still has a proper test, and the cross-section is not it. The identifying variation the model calls for is within-occupation task recomposition over time, whether the paid content of exposed bundles is redrawn in the direction the operator predicts. That test needs within-occupation microdata of the kind pioneered by Atalay et al. (2020), and we leave it to future empirical work. The null in the deployment window is not evidence against the model either. The model does not predict that two years of diffusion should be visible in occupational median wages, and a strong pass there would itself have been suspect.

The exercise carries a wider caution. A naive correlation between AI exposure and 2019–2025 occupational wage growth has the right sign, a small p-value, and nothing to do with AI. It measures the pandemic compression. This is the exposure-versus-incidence distinction of Section 4.1 in empirical form. Technical exposure correlates with wage levels, wage levels drove relative wage growth in this period, and the chain of correlations manufactures apparent support for any mechanism whose cross-sectional footprint is a wage gradient, strongest where that footprint is steepest.

9 Discussion

The central contribution of the paper is to locate automation and reinstatement in a priced task space. Reinstatement is not an aggregate return of labour to production. It is a spatial process in which new tasks must survive capital and then bind to existing capabilities. The same amount of task creation can therefore have different consequences depending on where it appears.

This yields a distributional result that one-dimensional models cannot state. Reinstatement returns human task mass wherever it lands. It returns pay only where the price field rewards the restored work. New tasks in socio-cognitive directions can restore both work and pay. New tasks in manual or service directions may restore work without restoring pay.

Calibrated to task-level AI exposure, the mechanism gives the paper's quantitative answer. Automation cuts the labour share from one to 0.65, and reinstatement leaves it at 0.644, because 68 percent of the seeded mass binds to no occupation. The wage burden is carried by stripped bundle content (-0.35 log points, employment-weighted); the congestion that re-sorting induces nets to almost nothing in the mean and redistributes a tenth of a log point across occupations. The inference is bounded: the firm foundings are consistent with the seeding prediction, the 2019–2025 wage cross-section cannot identify the wage prediction, and the robot wave's own window supports it, surviving a manufacturing-share control, without separating import exposure within the manufacturing arc. The results stand as a model-internal account that the data are consistent with. The causal test remains within-occupation: whether exposed bundles shed and gain paid content along the directions the operator predicts, measured in microdata of the kind pioneered by Atalay et al. (2020).

The model also clarifies the Turing-trap logic (Brynjolfsson 2022). Default automation strips high-priced task content from occupational bundles. Default reinstatement often refills them with tasks that are easier to bind because they demand no new priced capabilities. The occupation remains, but its paid content declines.

The unbound tail is a watch region. Much of the seeded mass the model keeps unbound lies under the outer western tail of the seeding ring, where the robotics firms already sit and where no existing occupation reaches. The model makes the forward question precise: if boundary seeding is right, the coming years should deposit new firms and new occupational tasks under that tail before existing occupations can carry them. The startup cross-section is a first reading of a process that has barely begun.

The demand channel is distinct from the wage channel. Automation lowers unit cost where capital replaces labour, and the output market decides whether the saving expands or contracts employment. Elastic demand can turn labour-saving technology into employment growth; inelastic demand turns it into labour release. But the adoption margin limits this channel, because savings vanish where capital just breaks even with labour. The demand channel is therefore weak exactly where automation begins.

The model keeps exposure and incidence separate. Technical exposure is what a technology can do; economic incidence is what it is paid to do. A weak raw correlation between exposure and wages does not refute the mechanism, because adoption depends on exposure relative to price. The price field turns exposure into incidence.

Several limitations bound the results, and each is a stated primitive rather than an oversight. The comparative statics hold the readiness fields e_o fixed between the two rest points: capability acquisition is treated as slower than the shock, a timescale separation in the spirit of Adão, Beraja, and Pandalai-Nayar (2024). The attachment capacity $C(\mathbf{r})$ is linear in employment. The output market is partial: each place faces independent isoelastic demand, with no cross-place substitution or income effects. The price field is held fixed under the technology shock, even though it can be interpreted as an assignment-equilibrium object. Re-solving the assignment after the shock is a first-order extension that should dampen displacement by allowing the price of crowded or cheapened work to adjust; the present paper deliberately holds $\Pi(\mathbf{r})$ fixed to isolate the spatial automation and reinstatement mechanisms.

The model is also static: it compares two rest points, and unbound tasks are read off as a density rather than accumulated as a stock. The fate accounting suggests that this is the most promising continuation of the framework. If unbound mass accumulates while occupations absorb new work at a finite pace, the speed of the shock relative to the speed of absorption becomes an economic variable in its own right, and the destination of reinstated work — which occupations end up holding it — may depend on the path and not only on the endpoint. We regard a dynamic treatment built on these observations as interesting future work. The firm positions make this concrete. AI and robotics firms occupy the bindable part of the seeded region, and the unbound field extends radially beyond them (Figure 7). One reading is that the outer band holds seeded work not yet organised into occupations, latent room for new firms and new kinds of work. A cross-section cannot tell that apart from an outer band that stays sparse because its work resists binding, which is the distinction a stock-based dynamic treatment would draw.

The wage wedge also opens a connection to rent-based accounts of automation (Acemoglu and Restrepo 2026). If the residual wedge is interpreted as an occupation- or family-level rent, the same price gate implies that high-wedge work is more likely to cross the adoption margin. The within-group compression implications of that extension require rent dispersion within exposed groups and are left to separate work.

The main implication is simple. The question is not whether automation creates new tasks. The question is where those tasks appear, whether labour keeps them, and whether the workers who can bind them already hold the priced capabilities they require.

A The assignment programme

This appendix specifies the assignment equilibrium behind Section 3.3 and reports the test sequence that selects it. All quantities are computed on the frozen inputs of Section 6.1; no observed wage enters any computation, so agreement with the estimated field of eq. (25) is a test of the interpretation, not a fit to its target.

The field is not a simple scarcity price. If tasks were priced by a CES aggregator under uniform demand, price would fall with output density; in the data it barely responds ($\partial \ln \Pi / \partial \ln n_0 = -0.03$, an implied substitution elasticity of about 35), and the dear socio-cognitive arc is the densely staffed one. Nor are the capability coefficients ordinary factor prices: they do not rise with capability scarcity, and one is negative once the others are held fixed, so they read as hedonic coefficients rather than prices of factors in fixed supply. Log price is, however, almost exactly linear in priced capability content ($R^2 = 0.96$). The field is therefore better read as an assignment price in the sense of multidimensional sorting (Lindenlaub 2017): the shadow price of matching heterogeneous capability bundles to heterogeneous task requirements under directionally biased demand.

The numerical assignment closure makes this concrete. Occupations are capability types whose productivity at a task location is their readiness in priced capabilities, the same object that later governs task attachment (Section 4.4); task demand is CES over locations, and each occupation allocates its employment toward locations that pay well relative to its readiness. No observed wage enters the computation, so agreement with the estimated field is a test rather than a fit. The comparison statistic is the directed gradient: the difference in median $\ln \Pi$ between a deep socio-cognitive reference arc and a deep technical–service reference arc (below); the estimated field puts it at +0.39. With uniform demand, the assignment recovers the shape of the reduced-form field (rank correlation +0.92) but produces a directed gradient of only +0.08. A single cognitive demand shifter,

$$\alpha(\mathbf{r}) = \exp(\lambda q_{S1}(\mathbf{r})),$$

steepens the gradient to +0.36 while preserving the shape (rank correlation +0.97 at $\lambda = 0.2$). A production-complementarity alternative concentrates labour in the north but makes it too cheap, because output expands faster than price: the correlation between wages and the price surface turns negative. The dearness of cognitive work is therefore better understood as demand-driven assignment pressure than as productivity superiority.

The implication for this paper is limited but important. The price field can be treated as an equilibrium object: a capability-assignment cost surface with a demand-driven cognitive premium. The comparative statics below hold that surface fixed. Re-solving the assignment after a technology shock is a general-equilibrium extension and is expected to dampen, not reverse, the mechanisms studied here.

The programme

Task locations are the cells of a polar grid over the disk, with baseline task-work density $n_0(\mathbf{r})$ built from occupational bundles and employment. Occupations are capability types. The productivity of occupation o at location $\mathbf{r}$ is its readiness $e_o(\mathbf{r})$ of eq. (12), evaluated at the calibrated acquisition scale: the same primitive that governs task attachment in the main model governs comparative advantage here.

Demand for task output is CES over locations with substitution elasticity $\sigma = 3$ and location weights $\alpha(\mathbf{r})$, uniform in the baseline. With aggregate output $\bar{Y} = \left(\sum_{\mathbf{r}} \alpha\, y^{\rho}\right)^{1/\rho}$,

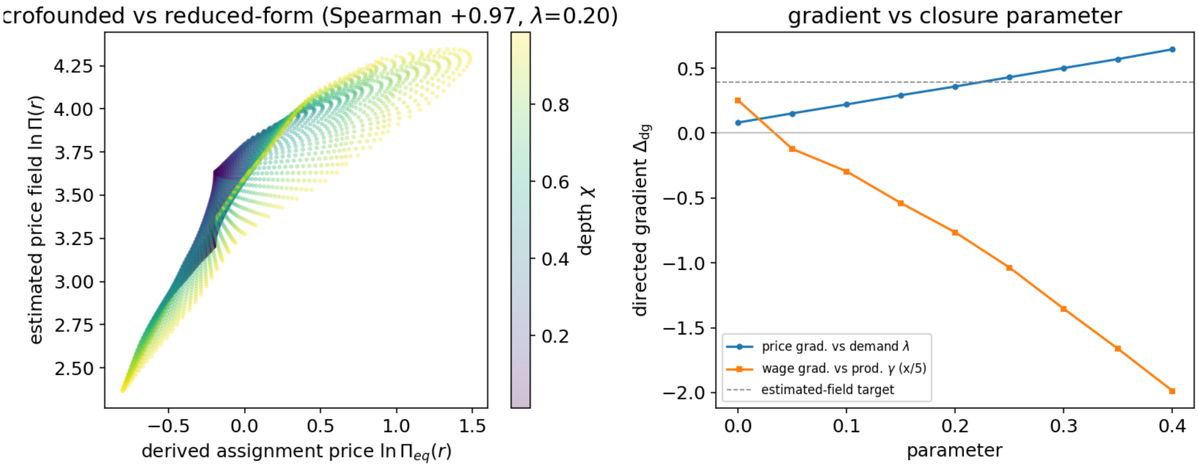


Figure 14: Assignment interpretation of the price field. The calibrated assignment price reproduces the reduced-form field closely when demand is biased toward cognitively intensive tasks.

$\rho = (\sigma - 1)/\sigma$, the inverse demand at a location is

$$\Pi_{\mathrm{eq}}(\mathbf{r}) = \alpha(\mathbf{r}) \left(\frac{\bar{Y}}{y(\mathbf{r})} \right)^{1/\sigma},$$

normalized to mean one as numeraire. Each occupation allocates its employment over locations by a logit in the payoff $\Pi_{\mathrm{eq}}(\mathbf{r})\, e_o(\mathbf{r})$ with smoothness κ, set to half the mean price. Output at a location is the readiness-weighted employment allocated there, $y(\mathbf{r}) = \sum_o L_o\, s_o(\mathbf{r})\, e_o(\mathbf{r})$, and the fixed point of prices and allocations is found by damped iteration. Occupation earnings per worker are $w_o = \sum_{\mathbf{r}} s_o(\mathbf{r})\, \Pi_{\mathrm{eq}}(\mathbf{r})\, e_o(\mathbf{r})$.

The comparison statistic used throughout is the directed gradient Δ_{dg}: the difference in median $\ln \Pi$ between a deep socio-cognitive reference arc ($\cos\xi > 0.5$, $\chi > 0.4$) and a deep technical–service reference arc ($\cos\xi < -0.3$, $\chi > 0.3$). On the estimated field of eq. (25) the directed gradient is $+0.39$; on log employment density it is $+0.71$, so the dear side of the disk is also the densely staffed side.

The test sequence

T1, task scarcity: rejected. If the field were a CES scarcity price under uniform demand, $\ln \Pi$ would fall in log density with slope $-1/\sigma$. The estimated slope is $\partial \ln \Pi / \partial \ln n_0 = -0.028$ (rank correlation -0.12), an implied elasticity of about 35, and the directed density gradient of $+0.71$ puts the mass where the price is high. Scarcity does not generate the field.

T2, capability pricing: confirmed. Log price is nearly linear in priced capability content: $\ln \Pi \sim a + b \sum_k v_k q_k(\mathbf{r})$ fits with $R^2 = 0.956$. The field is a hedonic capability surface with S1 dominant.

T3, capability scarcity: rejected. If the weights v_k were factor prices of capabilities in fixed supply, they would rise with scarcity. Across clusters, the correlation between v_k and inverse aggregate supply is -0.28, and the unrestricted A1 coefficient is negative (-0.030). The weights are hedonic coefficients, and pricing them requires an assignment rather than a factor market.

T4, assignment under uniform demand: shape. The equilibrium price of the programme above, computed without wages, reproduces the estimated field at rank correlation $+0.92$, robust to the allocation smoothness ($+0.90/+0.92/+0.92$ at κ equal to 0.2, 0.5, and 1.0 times the mean price). Its directed gradient is $+0.08$: the assignment carries the shape but only a quarter of the gradient.

T5, cognitive demand bias: closure. The demand shifter $\alpha(\mathbf{r}) = \exp(\lambda\, q_{S1}(\mathbf{r}))$, with q_{S1} standardized, is swept over $\lambda \in \{0.10, 0.20, 0.25, 0.30, 0.40\}$. At $\lambda = 0.2$ the directed gradient reaches $+0.36$ at rank correlation $+0.97$, and the employment gradient turns weakly positive ($+0.02$), matching the sign of the data. One parameter closes the gap that uniform demand leaves.

T6, productivity complementarity: rejected. The alternative multiplies productivity by a complementarity between occupation and task capability content. It concentrates employment in the north (directed employment gradient up to $+0.6$) but crashes its price: output expands faster than demand absorbs it, and the correlation between occupation wages and the price surface turns negative (-0.25 to -0.34 across the sweep). Abundant-but-cheap is the wrong sign for the data.

One residual remains. The magnitude of the north's employment concentration ($+0.71$ in the data) is not reproduced by the demand bias ($+0.02$ at $\lambda = 0.2$); matching it would require a richer demand structure. The interpretation the paper carries is therefore qualified: the field behaves as the price surface of a capability assignment with a demand-driven cognitive premium, and the premium's level, not the concentration of employment, is what the closure delivers.

B Price-field and geometry robustness

The price field of Section 3 is estimated once on the pooled cross-section. Figure 15 re-estimates its directional depth return sector by sector on the frozen inputs and compares it to the published sectoral estimates of Storck and Andersson (2026): the directional structure the main text uses is not an artefact of pooling.

C Notation

Symbols are collected below, grouped by role.

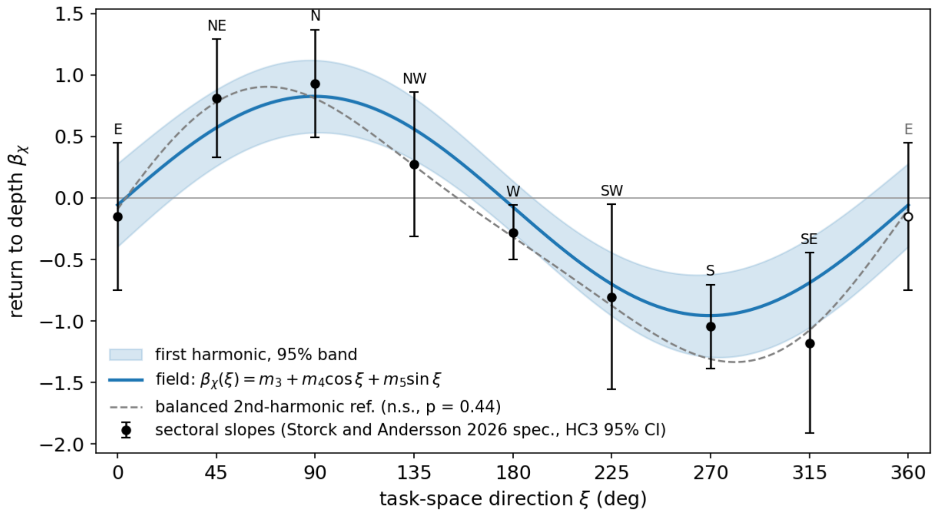


Figure 15: Directional return to depth. The axis variable $\beta_\chi(\xi) = m_3 + m_4 \cos\xi + m_5 \sin\xi$ is the estimated return to depth in direction ξ, the depth bracket of eq. (3), re-estimated on frozen inputs and compared to the sectoral estimates of Storck and Andersson (2026).

Symbol	Meaning
Task space and the price field	
$\mathbf{r} = (\xi, \chi)$	point in task space; ξ angular, χ radial coordinate ($\chi = 0$ centre, rim $\chi = 1$)
$\mu_o = \int \mathbf{r}\, b_o(\mathbf{r})\, d\mathbf{r}$	centroid of occupation o's task bundle b_o
$\Delta\mu_o$	shift of the centroid under the technology shock
$d(o, o') = \lVert \mu_o - \mu_{o'} \rVert$	task-space distance between occupations
$\Pi(\mathbf{r})$	price field: price of capability content at $\mathbf{r}$ (reduced form, estimated from wages)
$\nabla \ln \Pi$	gradient of the log price field
$\Pi_{\text{eff}} = e^{\omega_g}\Pi$	rent-adjusted effective price
$\widehat{\Delta \ln \Pi_o} = \Delta\mu_o \cdot \nabla \ln \Pi$	predicted directional wage pressure: first-order log-price change the shift implies (Section 8)
Capabilities	
$\mathcal{K}$, k	set of capability clusters and a generic cluster
$q_k(\mathbf{r})$, $q_{o,k}$	level of capability k at $\mathbf{r}$; occupation o's content
v_k	weight of capability k in the price functional (the gate weights, $v_k \geq 0$)

continued on next page

Symbol	Meaning
v_o, $\bar{v}$, $\bar{v}_k$	occupation o's capability profile and the mean profile (eq. (1)); $\bar{v}_k$ the cluster plane mean (eq. (2)). Unrelated to the gate weights v_k despite the shared letter
p_k, κ	per-capability price and intercept in $\ln \Pi = \kappa + \sum_k p_k q_k$
Technology field and takeover	
K	a technology field (cognitive or robot)
z_K, A_K	reach and amplitude of the field (calibration)
s_K, $\phi_K(\mathbf{r})$	replacing share of the field's effectiveness ($s_K = 1$ all replacing, 0 augmenting); ϕ_K technical exposure at $\mathbf{r}$
$a(\mathbf{r})$	operated (takeover) share at $\mathbf{r}$
R, τ	automation cost threshold (takeover compares Π to $R/(s_K \phi_K)$) and margin width
D_o	displaced share of occupation o
$\iota_o(\mathbf{r})$, $B_o = \int \iota_o \, d\mathbf{r}$	reinstatement inflow and reinstated mass
γ	reinstatement fraction ($= 0.5$)
$\hat{g}(\mathbf{r})$, M, $\zeta(\mathbf{r}) = M\hat{g}(\mathbf{r})$	normalized boundary density (eq. (10)), seeded task mass $M = \gamma \Delta \Gamma^D$, and seeded density
$\delta_o(\mathbf{r})$, $\bar{\delta}_o(\mathbf{r})$	priced capability deficit (eq. (11)) and surplus of occupation o at $\mathbf{r}$
$e_o(\mathbf{r})$	readiness to attach (eq. (12))
$C(\mathbf{r}) = \sum_o L_o e_o(\mathbf{r})$	attachment capacity at $\mathbf{r}$
$u(\mathbf{r})$	unbound task density (eq. (17))
$c(\mathbf{r})$	blended unit cost after takeover (eq. (9))
β	congestion exponent in $V(\mathbf{r}) = \mathcal{D}\,\Pi\, n^{\beta}$; human task-work is paid its marginal value product out of βV, capital is paid its cost, $(1-\beta)V$ is the congestion rent to the fixed factor
$\Phi(C) = C/(1+C)$	bound fraction at congestion C
Δw_o	bundle wage change of occupation o (eq. (16))
Wages and rents	
$w_{\text{obs},o}$, $w_{\text{bundle},o}$	observed wage and model bundle wage of occupation o
$\omega_o = \ln w_{\text{obs},o} - \ln w_{\text{bundle},o}$	occupation rent (residual of log wage on priced content)
ω_g	family wage wedge: the family mean of ω_o
η	demand elasticity (Section 7.3)
Equilibrium and mobility	

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Symbol	Meaning
$n(\mathbf{r})$	total task-work density (eq. (18))
$H(\mathbf{r}) = \sum_o L_o h_o(\mathbf{r})$	human task-work density
$\mathcal{D}(\mathbf{r}) = (c/\Pi)^{1-\eta}$	demand multiplier (eq. (19)); distinct from the displaced share D_o
$V(\mathbf{r})$	place output (eq. (20))
W_o	occupation value (eq. (21)); the object workers sort on
Λ	labour share of variable task-work income (eq. (24))
ℓ	acquisition scale in the readiness e_o (eq. (12))
L_0, L	baseline and re-solved employment allocation
λ	cognitive demand-shifter exponent in the assignment closure (Appendix A)
c_m, ν	mobility cost and logit smoothness in the re-sorting map
α_o	anchoring constants in the re-sorting map; observed employment is the zero-technology fixed point
ρ, λ_{over}	attachment locality scale and over-qualification weight (eq. (12))

Data and code availability

All inputs, code, and frozen outputs are public at `https://github.com/JoakimStorck/technology-fields`. Every result-bearing number in the paper traces to a producing script in `scripts/`, with its output frozen in `results/`; the pre-registered placebo test records its hypotheses in the producing script's docstring.

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Appendix 1: Bundle examples with tasks

Figure 16 reproduces panel (b) of Figure 1 with every task numbered, and Table 8 lists the tasks behind the numbers.

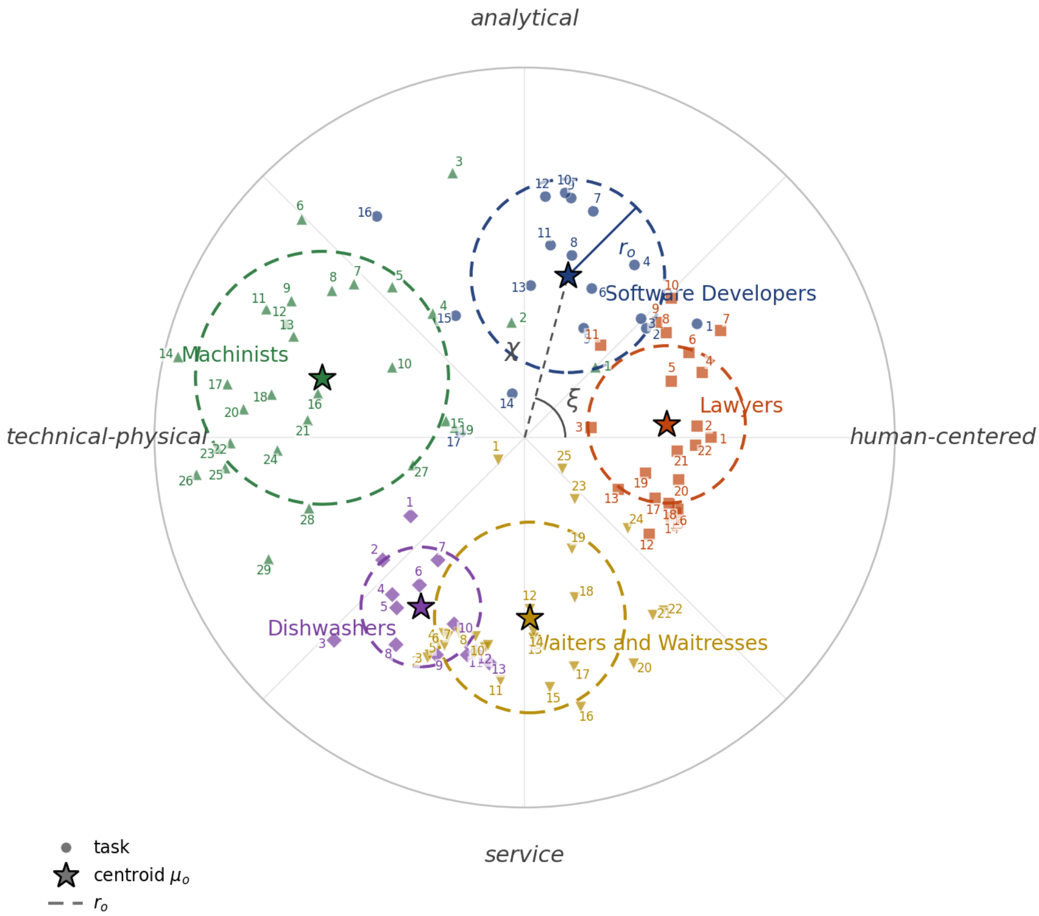

Figure 16: Panel (b) of Figure 1 with each task numbered. Within each occupation the numbers run from one around the disk in order of angle and key into Table 8.

Table 8: Tasks of the example occupations of Figure 1, keyed to the numbers in the supplementary numbered panel. Direction is the compass sector of the task on the disk.

No.	Task	Direction
Software Developers		

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Table 8 (continued)

No.	Task	Direction
1	Consult with customers or other departments on project status, proposals, or technical issues, such as software system design or maintenance.	human-centered
2	Prepare reports or correspondence concerning project specifications, activities, or status.	human-centered
3	Confer with data processing or project managers to obtain information on limitations or capabilities for data processing projects.	analytical
4	Confer with systems analysts, engineers, programmers and others to design systems and to obtain information on project limitations and capabilities, performance requirements and interfaces.	analytical
5	Supervise and assign work to programmers, designers, technologists, technicians, or other engineering or scientific personnel.	analytical
6	Supervise the work of programmers, technologists and technicians and other engineering and scientific personnel.	analytical
7	Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.	analytical
8	Obtain and evaluate information on factors such as reporting formats required, costs, or security needs to determine hardware configuration.	analytical
9	Develop or direct software system testing or validation procedures, programming, or documentation.	analytical
10	Analyze information to determine, recommend, and plan installation of a new system or modification of an existing system.	analytical
11	Determine system performance standards.	analytical
12	Design, develop and modify software systems, using scientific analysis and mathematical models to predict and measure outcomes and consequences of design.	analytical
13	Store, retrieve, and manipulate data for analysis of system capabilities and requirements.	analytical
14	Coordinate installation of software system.	analytical
15	Modify existing software to correct errors, adapt it to new hardware, or upgrade interfaces and improve performance.	analytical
16	Monitor functioning of equipment to ensure system operates in conformance with specifications.	analytical
17	Train users to use new or modified equipment.	technical-physical

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Table 8 (continued)

No.	Task	Direction
Lawyers		
1	Advise clients concerning business transactions, claim liability, advisability of prosecuting or defending lawsuits, or legal rights and obligations.	human-centered
2	Interpret laws, rulings and regulations for individuals and businesses.	human-centered
3	Prepare, draft, and review legal documents, such as wills, deeds, patent applications, mortgages, leases, and contracts.	human-centered
4	Confer with colleagues with specialties in appropriate areas of legal issue to establish and verify bases for legal proceedings.	human-centered
5	Gather evidence to formulate defense or to initiate legal actions by such means as interviewing clients and witnesses to ascertain the facts of a case.	human-centered
6	Evaluate findings and develop strategies and arguments in preparation for presentation of cases.	human-centered
7	Help develop federal and state programs, draft and interpret laws and legislation, and establish enforcement procedures.	human-centered
8	Study Constitution, statutes, decisions, regulations, and ordinances of quasi-judicial bodies to determine ramifications for cases.	human-centered
9	Analyze the probable outcomes of cases, using knowledge of legal precedents.	human-centered
10	Examine legal data to determine advisability of defending or prosecuting lawsuit.	human-centered
11	Search for and examine public and other legal records to write opinions or establish ownership.	analytical
12	Act as agent, trustee, guardian, or executor for businesses or individuals.	human-centered
13	Select jurors, argue motions, meet with judges, and question witnesses during the course of a trial.	human-centered
14	Present and summarize cases to judges and juries.	human-centered
15	Perform administrative and management functions related to the practice of law.	human-centered
16	Supervise legal assistants.	human-centered
17	Negotiate settlements of civil disputes.	human-centered

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Table 8 (continued)

No.	Task	Direction
18	Probate wills and represent and advise executors and administrators of estates.	human-centered
19	Negotiate contractual agreements.	human-centered
20	Prepare legal briefs and opinions, and file appeals in state and federal courts of appeal.	human-centered
21	Present evidence to defend clients or prosecute defendants in criminal or civil litigation.	human-centered
22	Represent clients in court or before government agencies.	human-centered
Machinists		
1	Advise clients about the materials being used for finished products.	analytical
2	Confer with engineering, supervisory, or manufacturing personnel to exchange technical information.	analytical
3	Test experimental models under simulated operating conditions, for purposes such as development, standardization, or feasibility of design.	analytical
4	Confer with numerical control programmers to check and ensure that new programs or machinery will function properly and that output will meet specifications.	analytical
5	Evaluate machining procedures and recommend changes or modifications for improved efficiency or adaptability.	analytical
6	Measure, examine, or test completed units to check for defects and ensure conformance to specifications, using precision instruments, such as micrometers.	technical-physical
7	Design fixtures, tooling, or experimental parts to meet special engineering needs.	technical-physical
8	Study sample parts, blueprints, drawings, or engineering information to determine methods or sequences of operations needed to fabricate products.	technical-physical
9	Calculate dimensions or tolerances, using instruments, such as micrometers or vernier calipers.	technical-physical
10	Program computers or electronic instruments, such as numerically controlled machine tools.	technical-physical
11	Operate equipment to verify operational efficiency.	technical-physical

continued on next page

Table 8 (continued)

No.	Task	Direction
12	Diagnose machine tool malfunctions to determine need for adjustments or repairs.	technical-physical
13	Support metalworking projects from planning and fabrication through assembly, inspection, and testing, using knowledge of machine functions, metal properties, and mathematics.	technical-physical
14	Dismantle machines or equipment, using hand tools or power tools to examine parts for defects and replace defective parts where needed.	technical-physical
15	Prepare working sketches for the illustration of product appearance.	technical-physical
16	Monitor the feed and speed of machines during the machining process.	technical-physical
17	Install experimental parts or assemblies, such as hydraulic systems, electrical wiring, lubricants, or batteries into machines or mechanisms.	technical-physical
18	Check work pieces to ensure that they are properly lubricated or cooled.	technical-physical
19	Dispose of scrap or waste material in accordance with company policies and environmental regulations.	technical-physical
20	Install repaired parts into equipment or install new equipment.	technical-physical
21	Establish work procedures for fabricating new structural products, using a variety of metalworking machines.	technical-physical
22	Set up or operate metalworking, brazing, heat-treating, welding, or cutting equipment.	technical-physical
23	Set up, adjust, or operate basic or specialized machine tools used to perform precision machining operations.	technical-physical
24	Maintain machine tools in proper operational condition.	technical-physical
25	Machine parts to specifications, using machine tools, such as lathes, milling machines, shapers, or grinders.	technical-physical
26	Fit and assemble parts to make or repair machine tools.	technical-physical
27	Separate scrap waste and related materials for reuse, recycling, or disposal.	technical-physical
28	Lay out, measure, and mark metal stock to display placement of cuts.	technical-physical

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Table 8 (continued)

No.	Task	Direction
29	Align and secure holding fixtures, cutting tools, attachments, accessories, or materials onto machines.	technical-physical
Dishwashers		
1	Clean garbage cans with water or steam.	technical-physical
2	Wash dishes, glassware, flatware, pots, or pans, using dishwashers or by hand.	technical-physical
3	Transfer supplies or equipment between storage and work areas, by hand or using hand trucks.	service
4	Clean or prepare various foods for cooking or serving.	service
5	Sweep or scrub floors.	service
6	Place clean dishes, utensils, or cooking equipment in storage areas.	service
7	Maintain kitchen work areas, equipment, or utensils in clean and orderly condition.	service
8	Load or unload trucks that deliver or pick up food or supplies.	service
9	Sort and remove trash, placing it in designated pickup areas.	service
10	Prepare and package individual place settings.	service
11	Stock supplies, such as food or utensils, in serving stations, cupboards, refrigerators, or salad bars.	service
12	Set up banquet tables.	service
13	Receive and store supplies.	service
Waiters and Waitresses		
1	Prepare checks that itemize and total meal costs and sales taxes.	technical-physical
2	Roll silverware, set up food stations, or set up dining areas to prepare for the next shift or for large parties.	service
3	Fill salt, pepper, sugar, cream, condiment, and napkin containers.	service
4	Remove dishes and glasses from tables or counters, and take them to kitchen for cleaning.	service
5	Perform cleaning duties, such as sweeping and mopping floors, vacuuming carpet, tidying up server station, taking out trash, or checking and cleaning bathroom.	service
6	Prepare tables for meals, including setting up items such as linens, silverware, and glassware.	service

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Table 8 (continued)

No.	Task	Direction
7	Prepare hot, cold, and mixed drinks for patrons, and chill bottles of wine.	service
8	Clean tables or counters after patrons have finished dining.	service
9	Perform food preparation duties, such as preparing salads, appetizers, and cold dishes, portioning desserts, and brewing coffee.	service
10	Garnish and decorate dishes in preparation for serving.	service
11	Stock service areas with supplies such as coffee, food, tableware, and linens.	service
12	Bring wine selections to tables with appropriate glasses, and pour the wines for customers.	service
13	Write patrons' food orders on order slips, memorize orders, or enter orders into computers for transmittal to kitchen staff.	service
14	Serve food or beverages to patrons, and prepare or serve specialty dishes at tables as required.	service
15	Take orders from patrons for food or beverages.	service
16	Escort customers to their tables.	service
17	Assist host or hostess by answering phones to take reservations or to-go orders, and by greeting, seating, and thanking guests.	service
18	Collect payments from customers.	service
19	Explain how various menu items are prepared, describing ingredients and cooking methods.	service
20	Inform customers of daily specials.	service
21	Present menus to patrons and answer questions about menu items, making recommendations upon request.	service
22	Provide guests with information about local areas, including directions.	service
23	Check with customers to ensure that they are enjoying their meals, and take action to correct any problems.	service
24	Describe and recommend wines to customers.	human-centered
25	Check patrons' identification to ensure that they meet minimum age requirements for consumption of alcoholic beverages.	human-centered